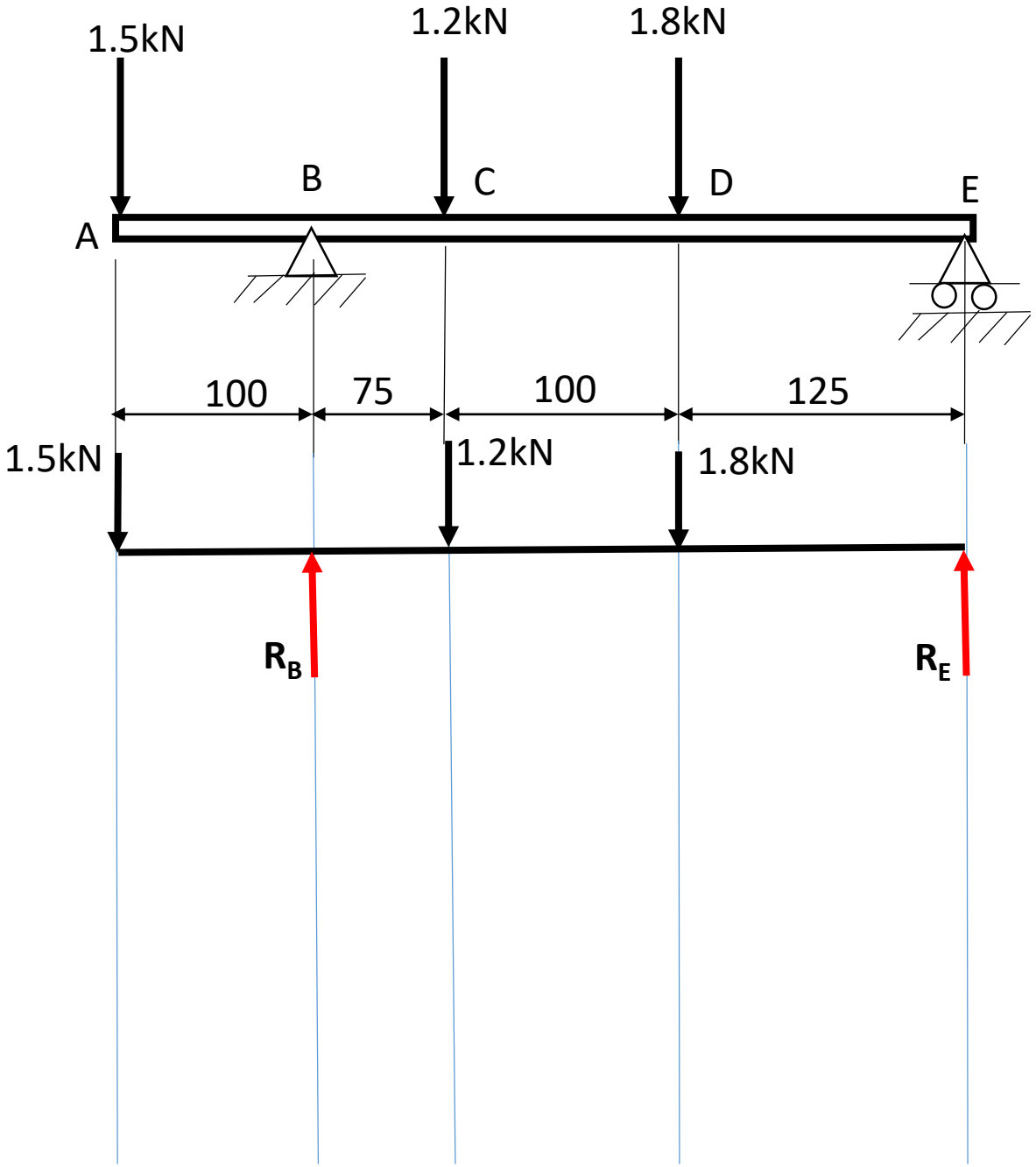


Kesit Tesirleri

Normal Kuvvet, Kesme Kuvveti ve Eğilme Momenti Diyagramlarının Çizim UYGULAMLARI

Dr. Bilçen MUTLU



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

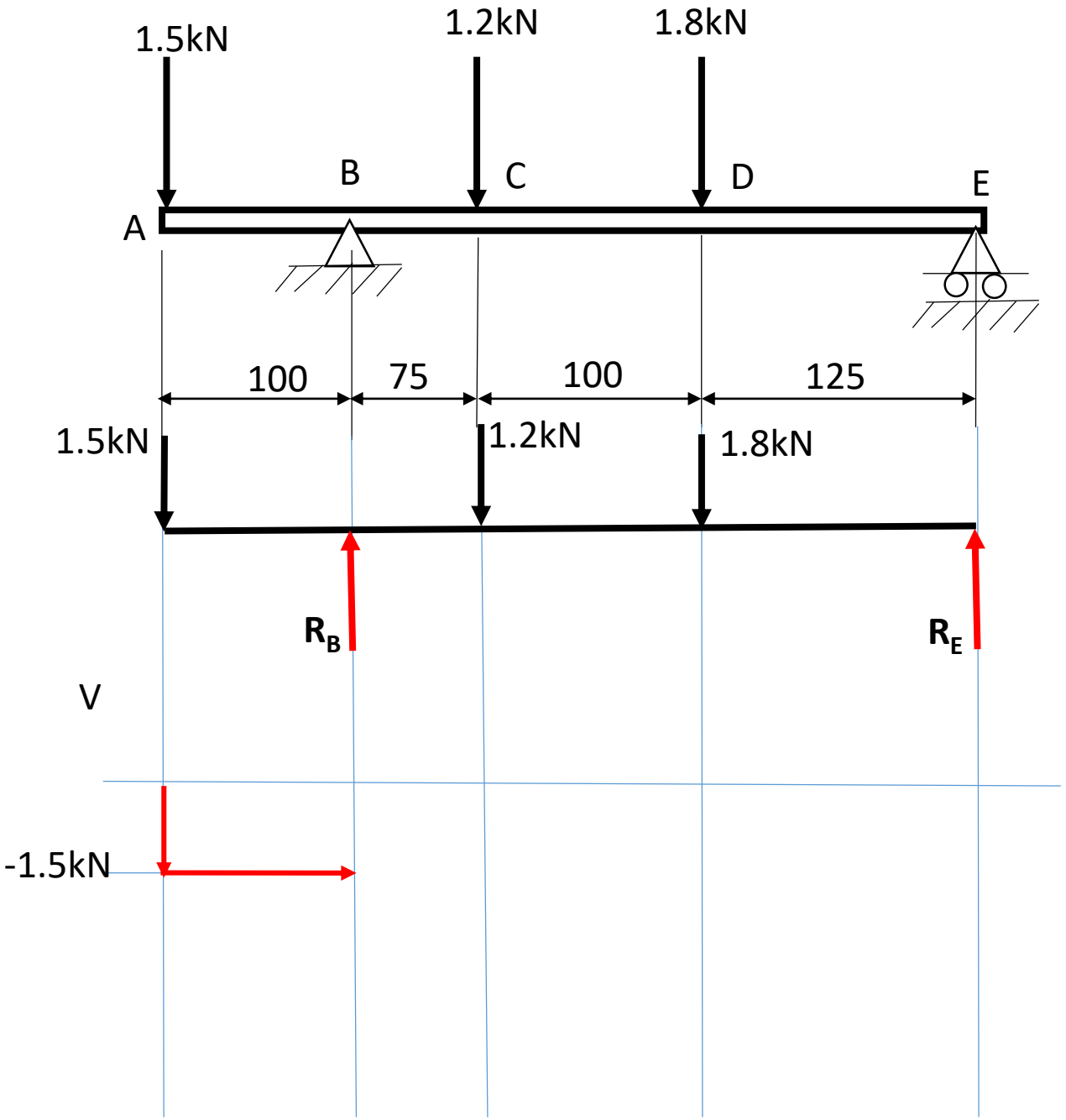
$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

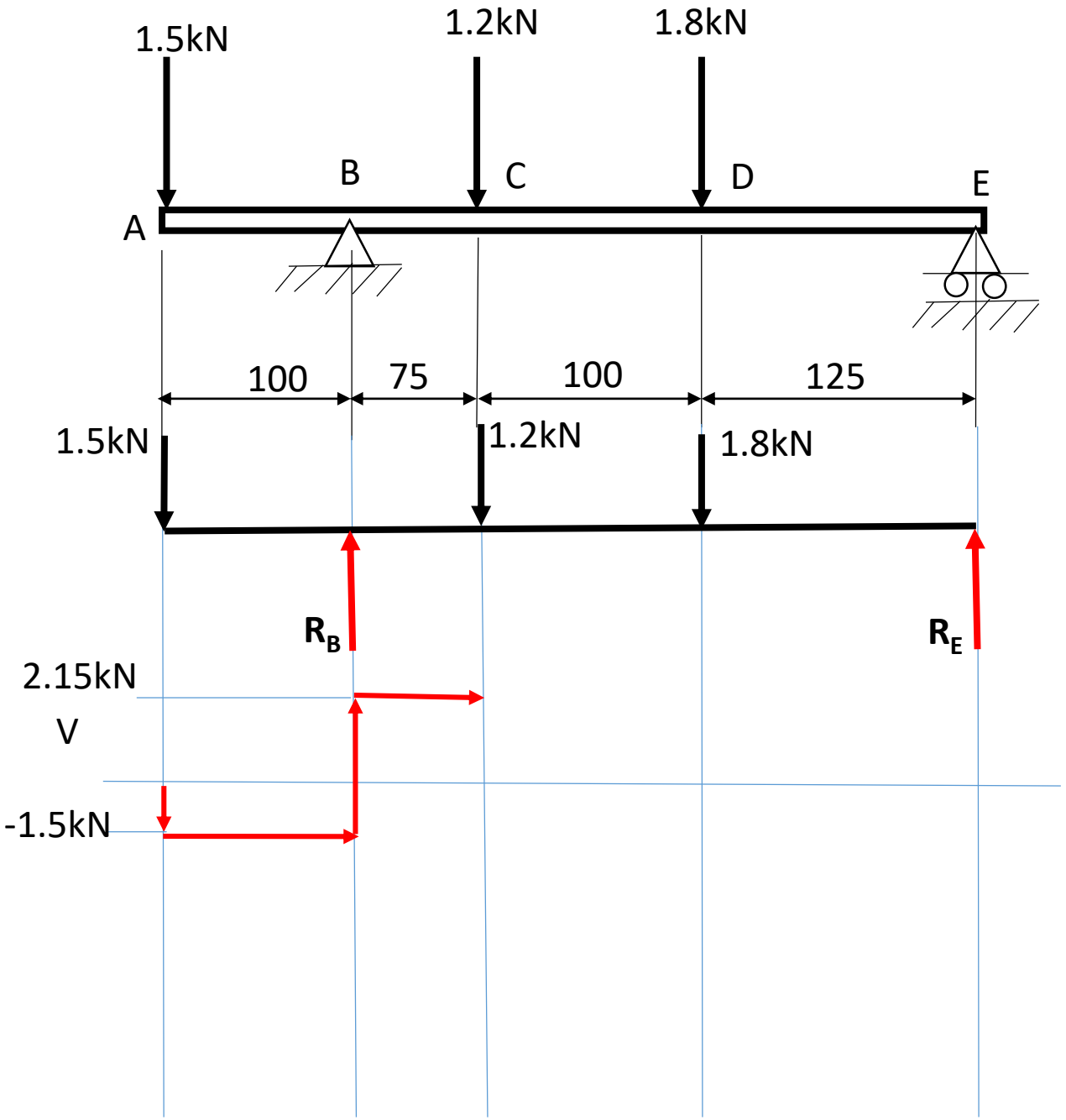
$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

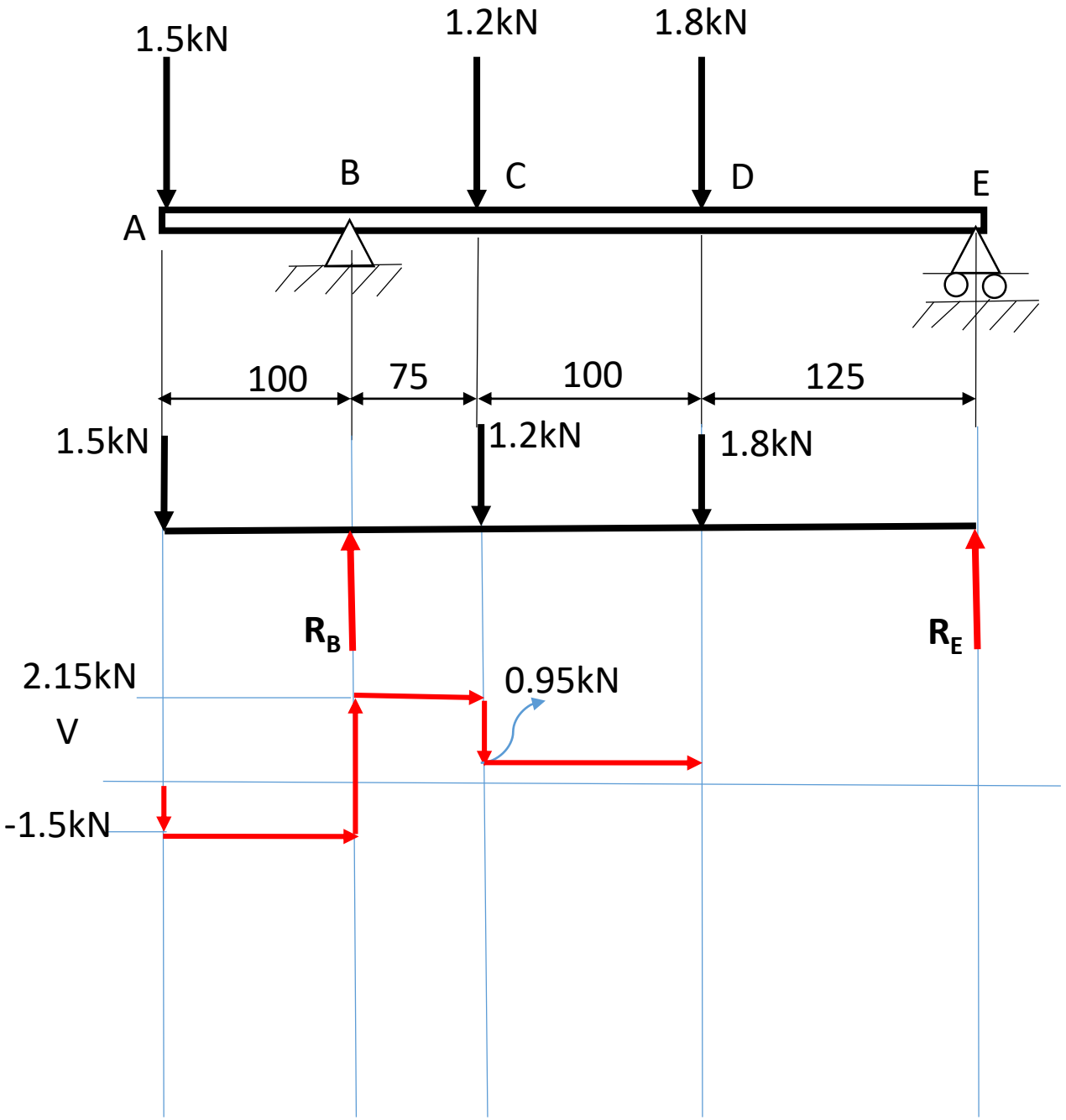
$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

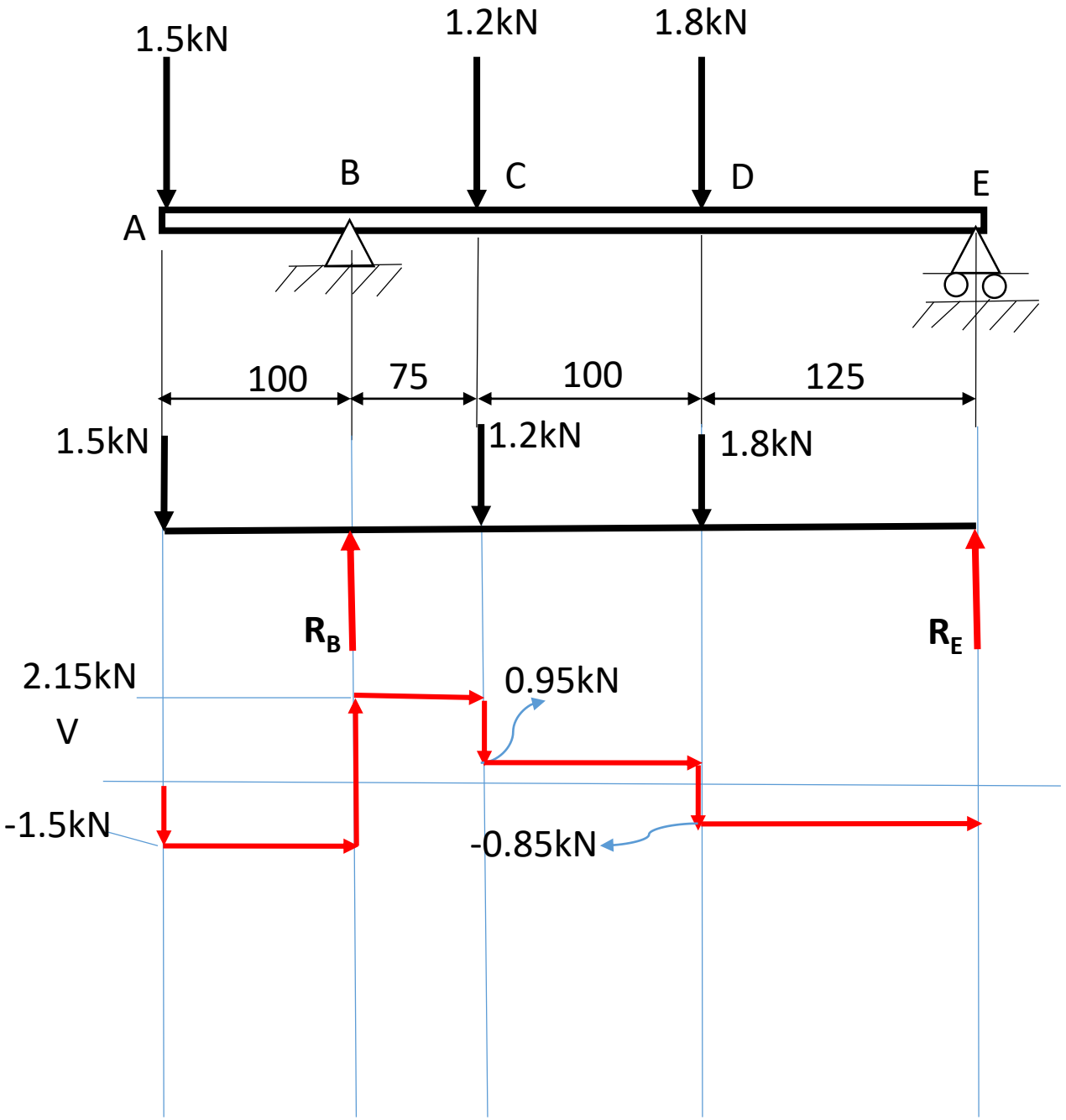
$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

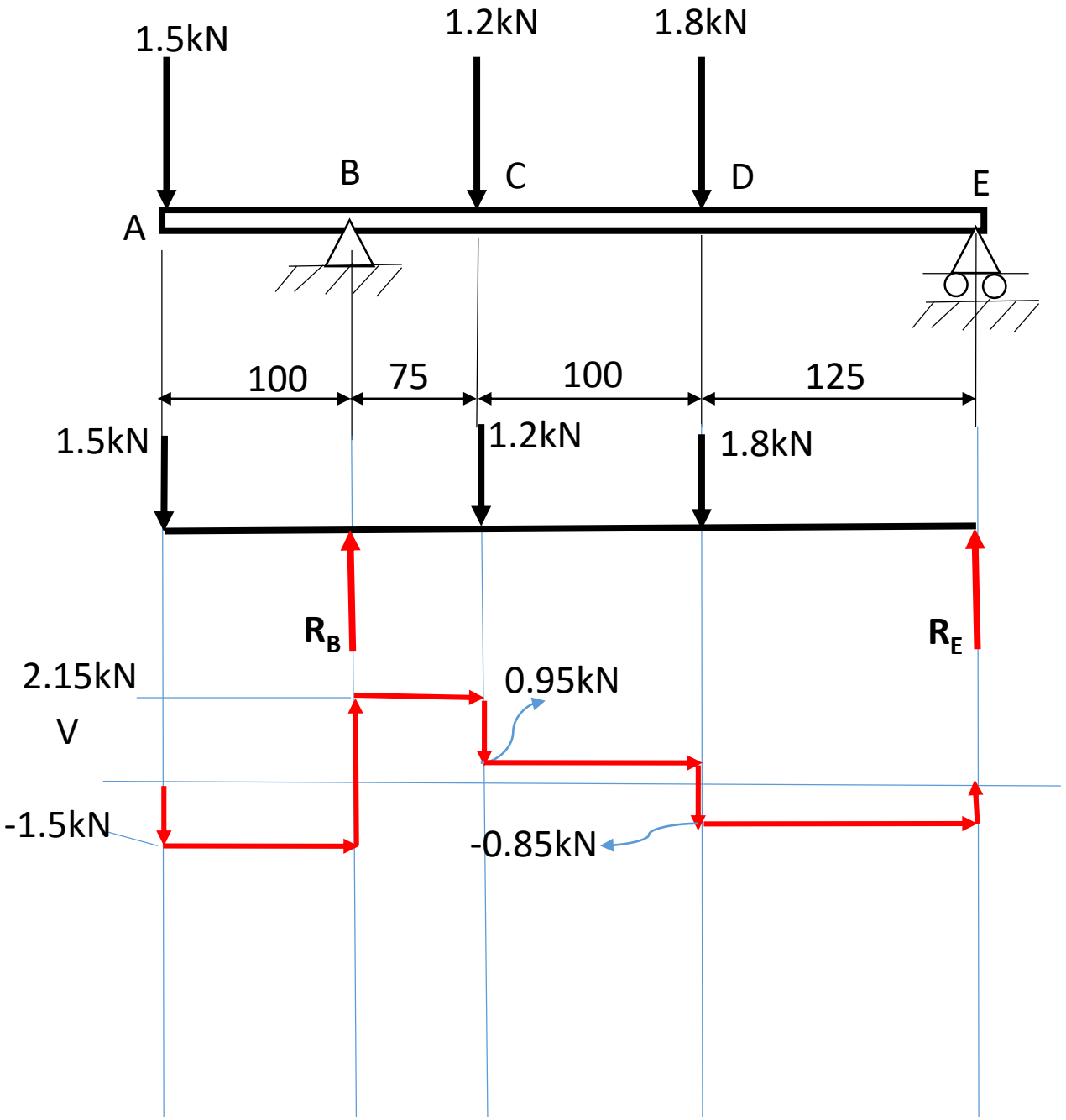
$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$



$$\downarrow_{+} \sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

$$\circlearrowleft_{+} \sum M_B = 0$$

$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

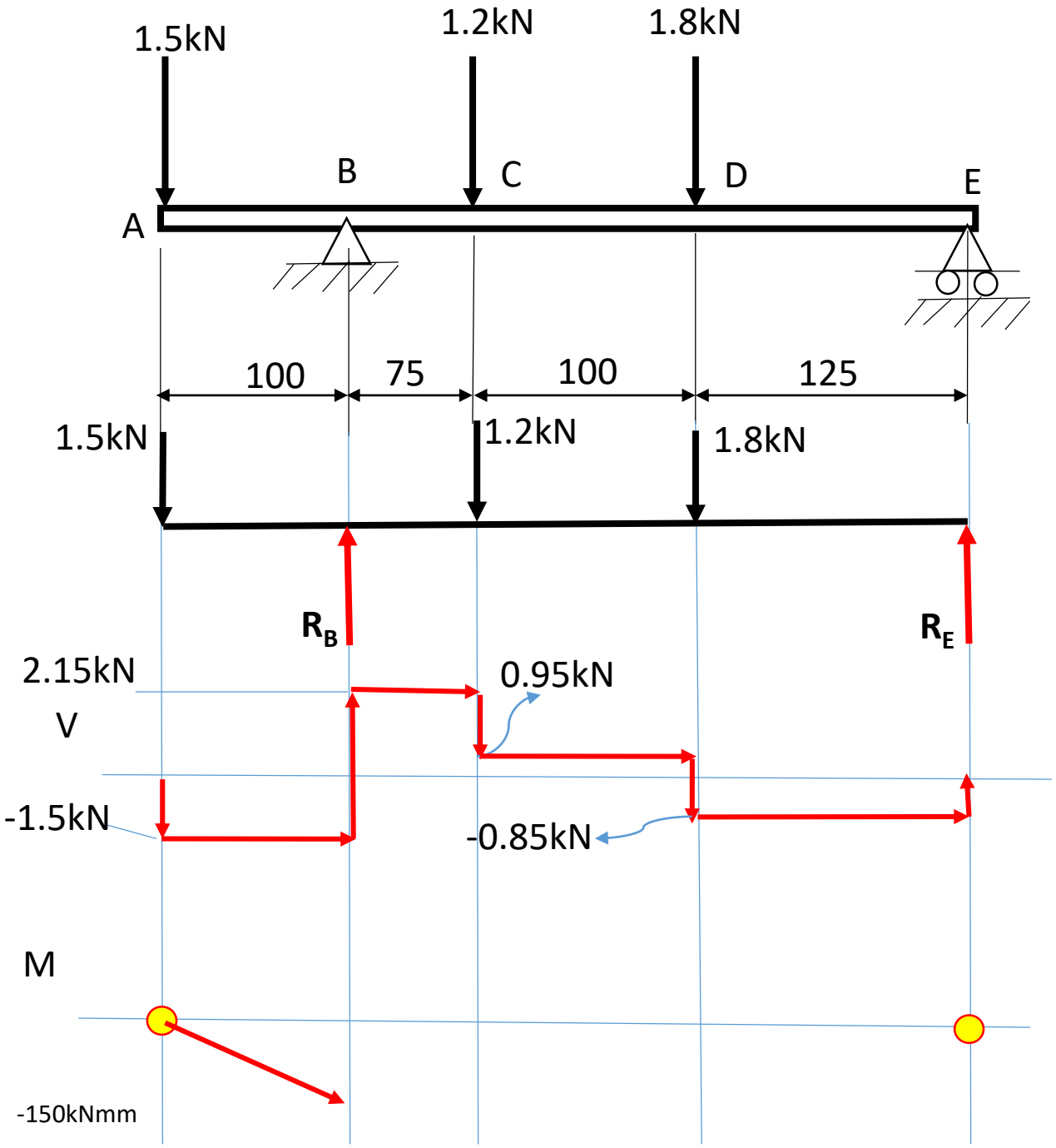
$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$

Kirişin kesme kuvveti diyagramı incelendiğinde B mesnedinde V_{max}=2.15 kN luk kuvvet ile kesilmeye zorlandığı görülmektedir.





$$\downarrow_{+}\sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

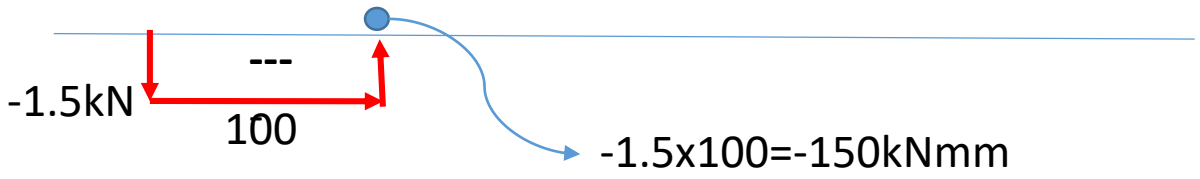
$$\circlearrowleft_{+}\sum M_B = 0$$

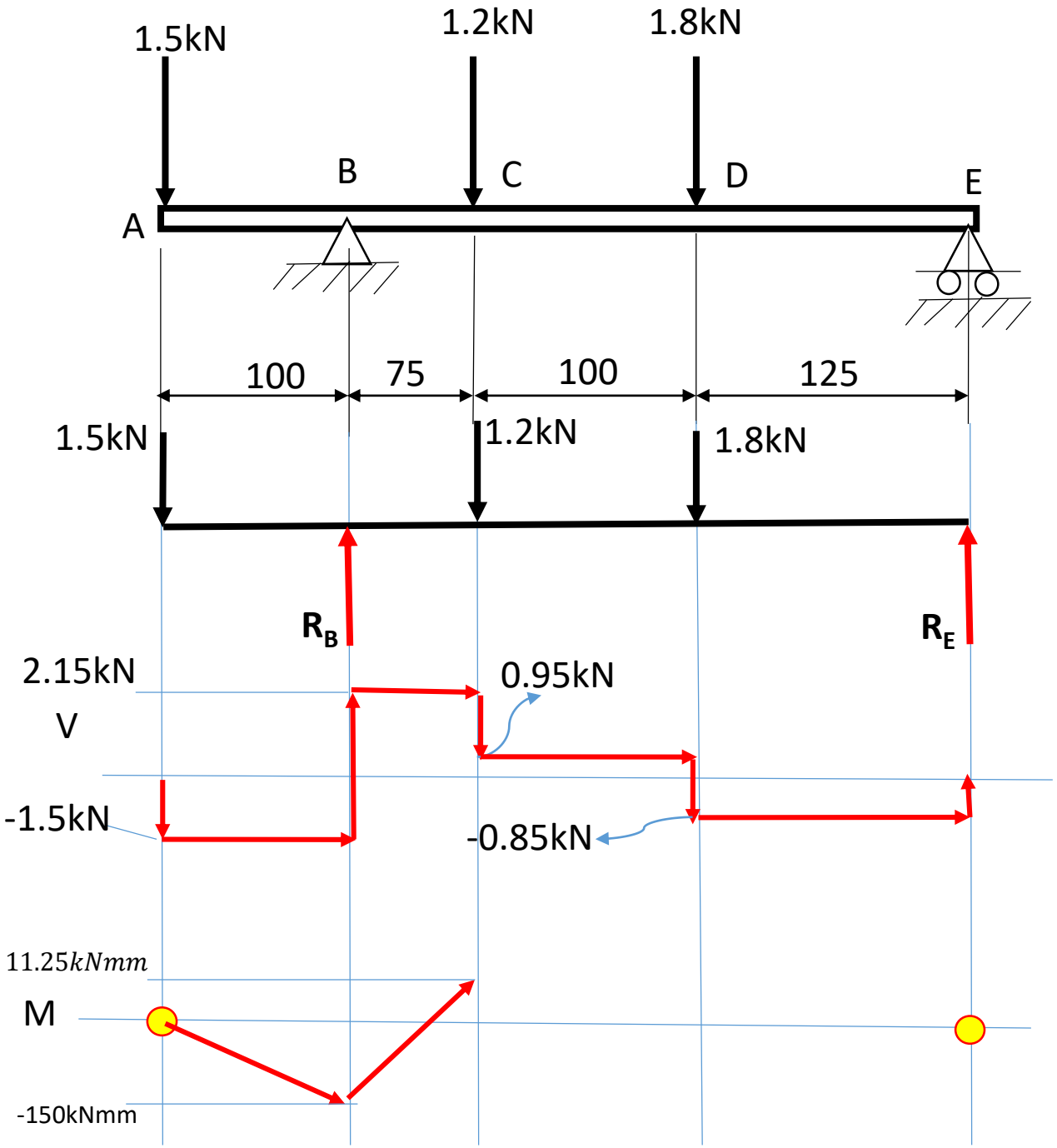
$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

$$R_B = 3.65\text{kN}$$





$$\downarrow_{+}\sum F_y = 1.5 - R_B + 1.2 + 1.8 - R_E = 0$$

$$R_B + R_E = 4.5$$

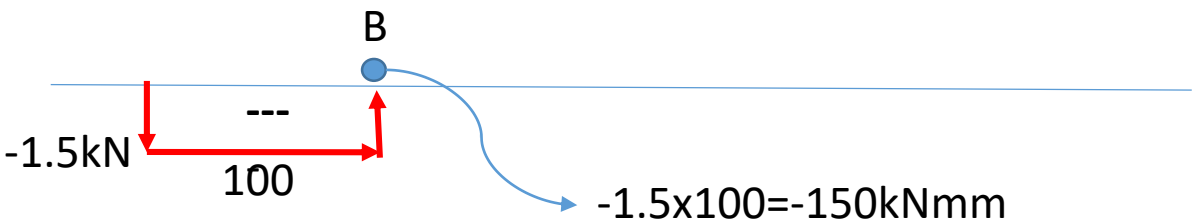
$$\curvearrowleft_{+}\sum M_B = 0$$

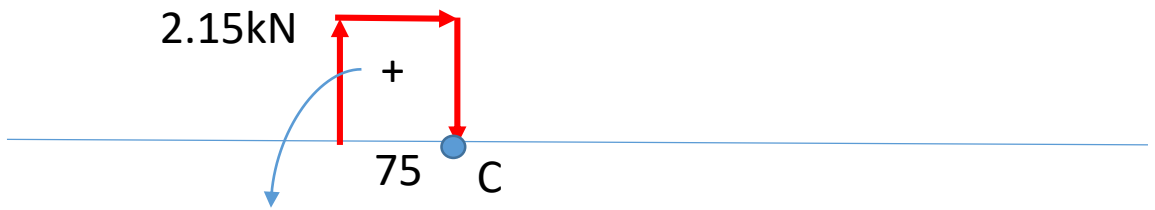
$$1.5x100 - 1.2x75 - 1.8x175 + R_E x300 = 0$$

$$R_E = 0.85kN$$

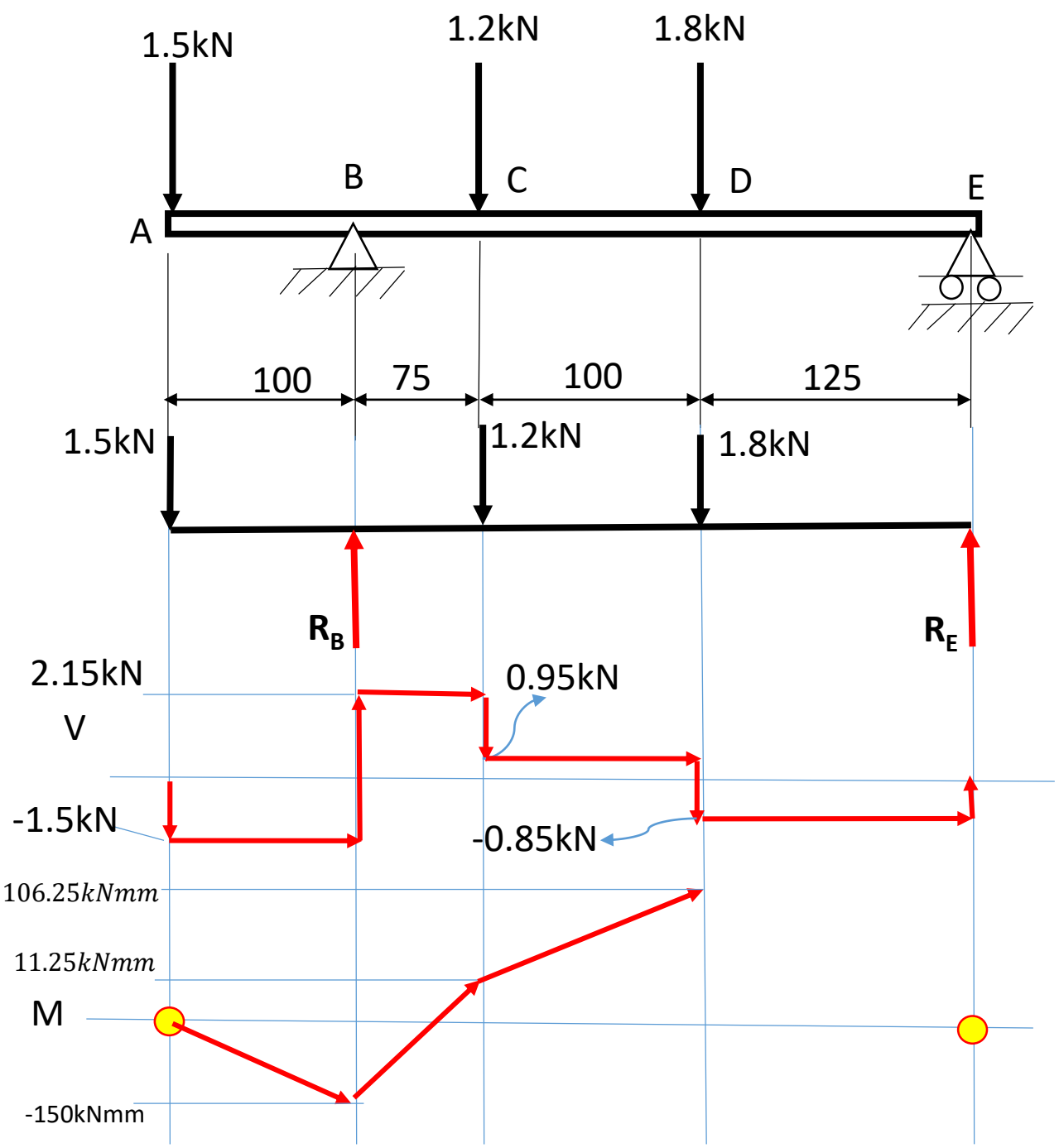
$$R_B + R_E = 4.5 \qquad R_B + 0.85 = 4.5$$

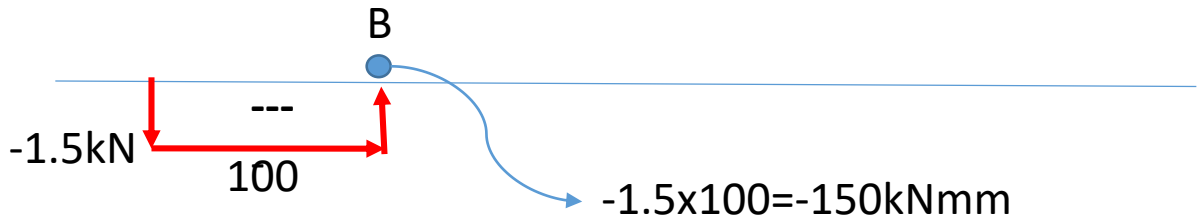
$$R_B = 3.65\text{kN}$$

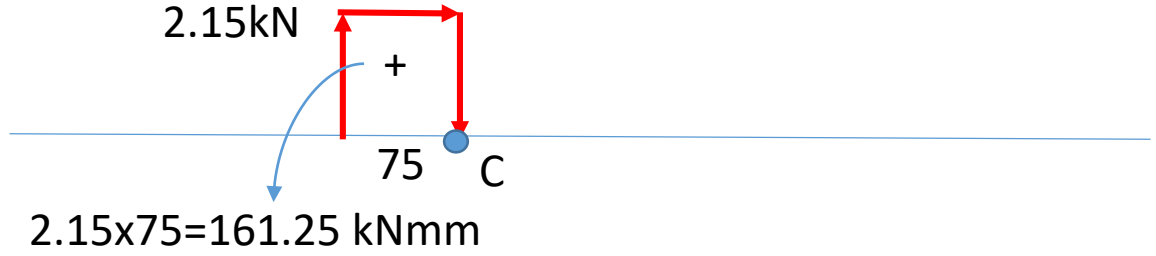




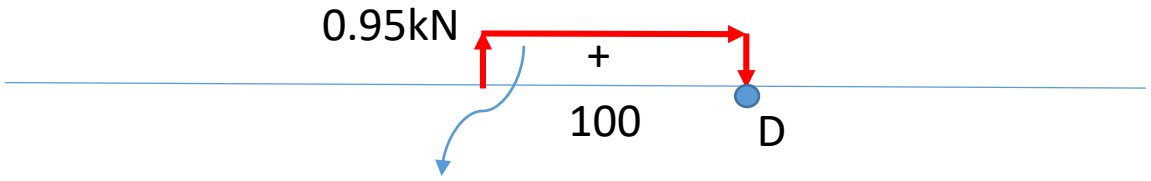
$$M_C = -150 + 161.25 = 11.25kNmm$$





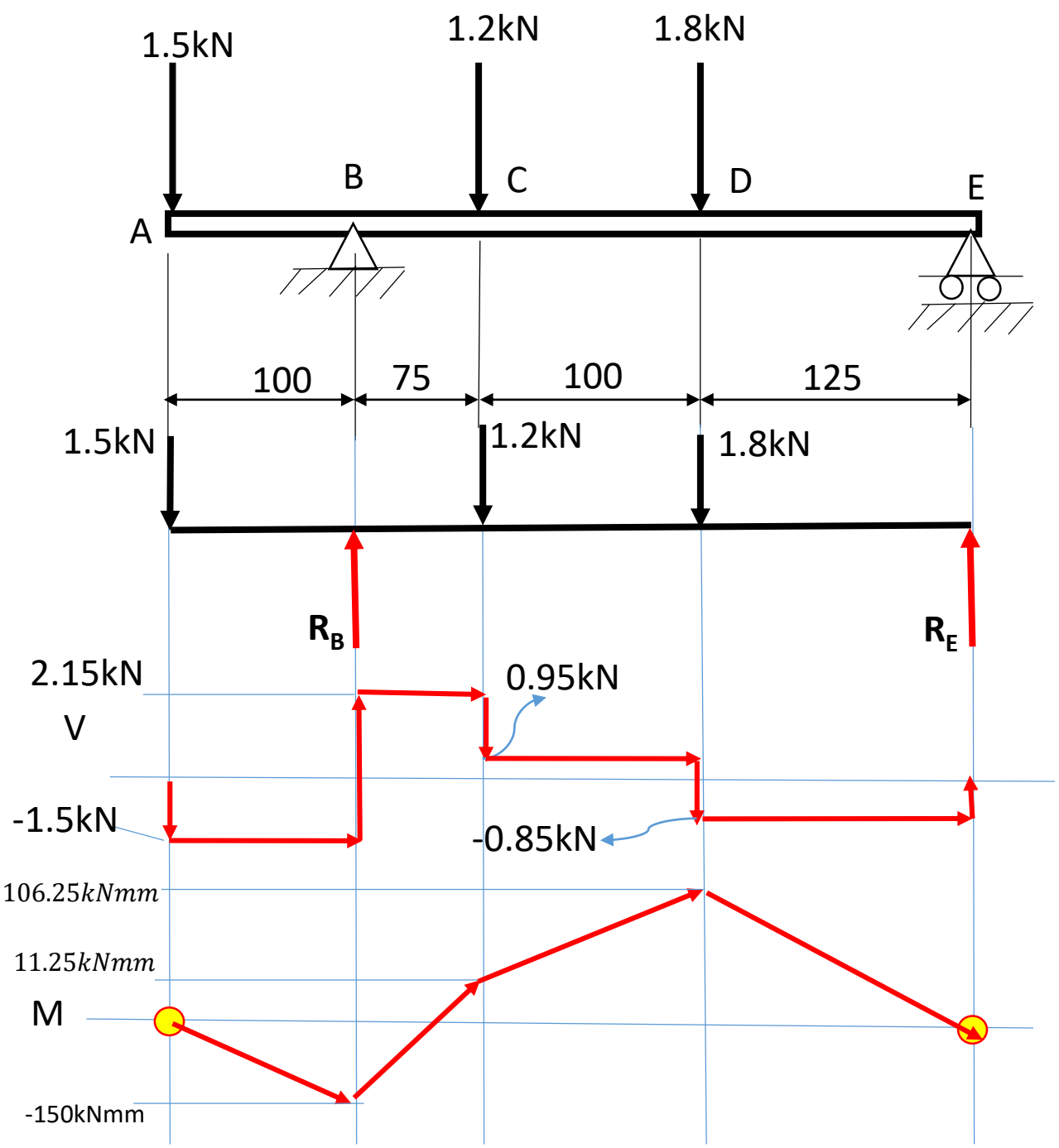


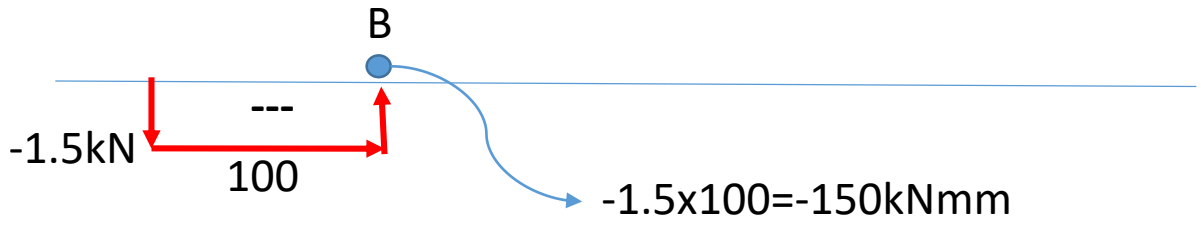
$$M_C = -150 + 161.25 = 11.25kNmm$$

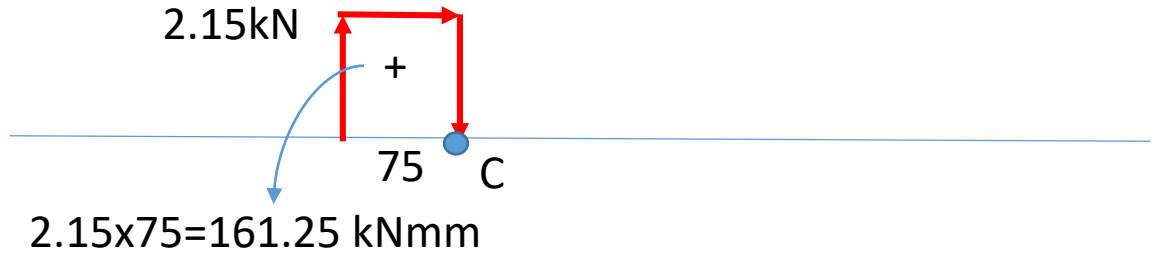


$$0.95x100 = 95kNmm$$

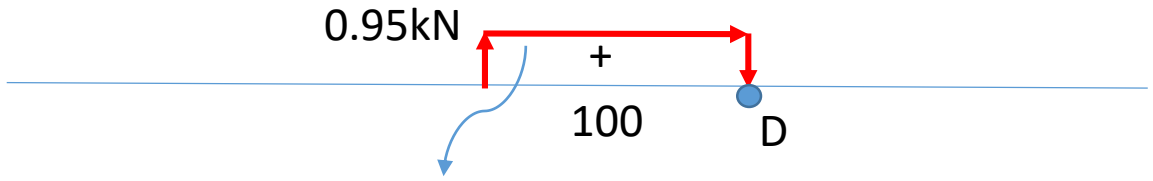
$$M_D = 11.25 + 95 = 106.25kNmm$$





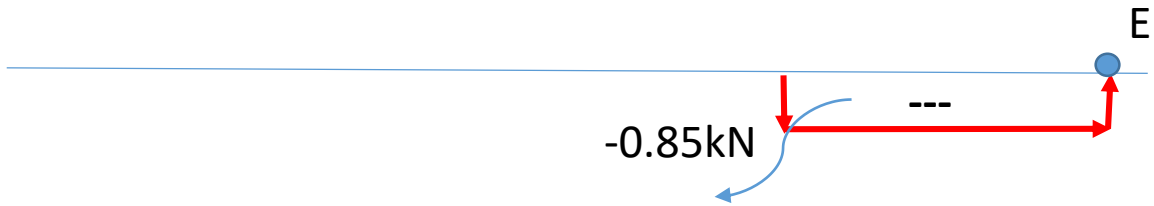


$$M_C = -150 + 161.25 = 11.25kNmm$$



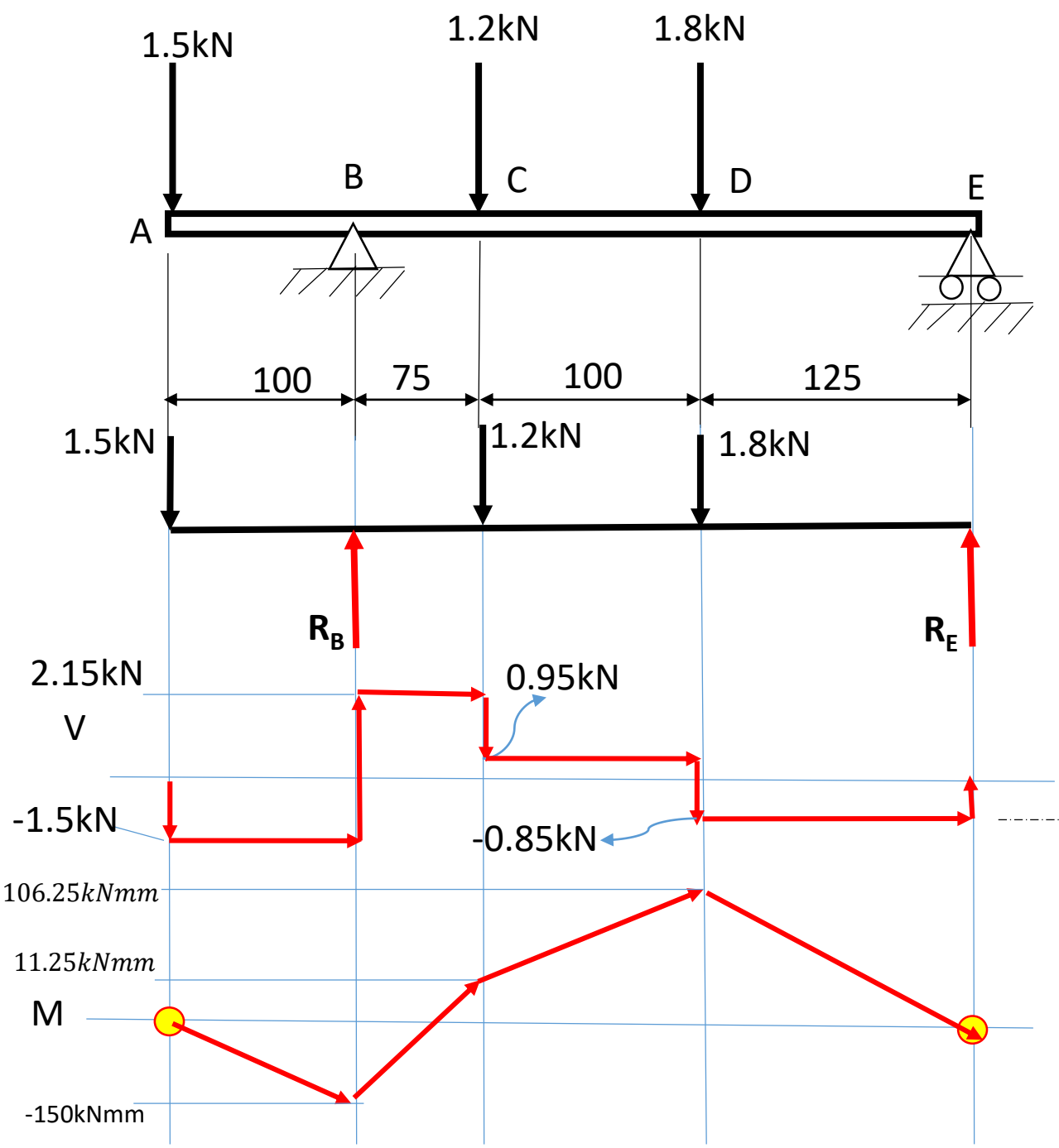
$$0.95x100 = 95kNmm$$

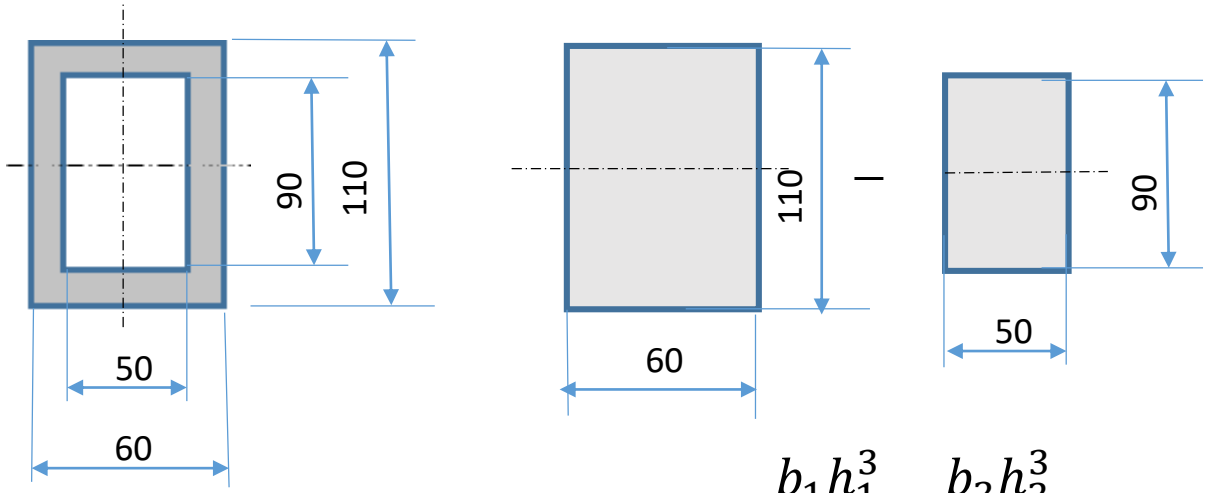
$$M_D = 11.25 + 95 = 106.25kNmm$$



$$0.85x125 = -106.25kNmm$$

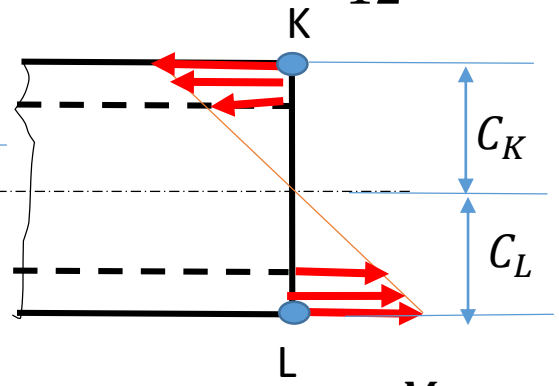
$$M_E = -106.25 + 106.25 = 0$$





$$I_G = \frac{b_1 h_1^3}{12} - \frac{b_2 h_2^3}{12}$$

$$I_G = \frac{60.110^3}{12} - \frac{50.90^3}{12} = 3617500mm^4$$

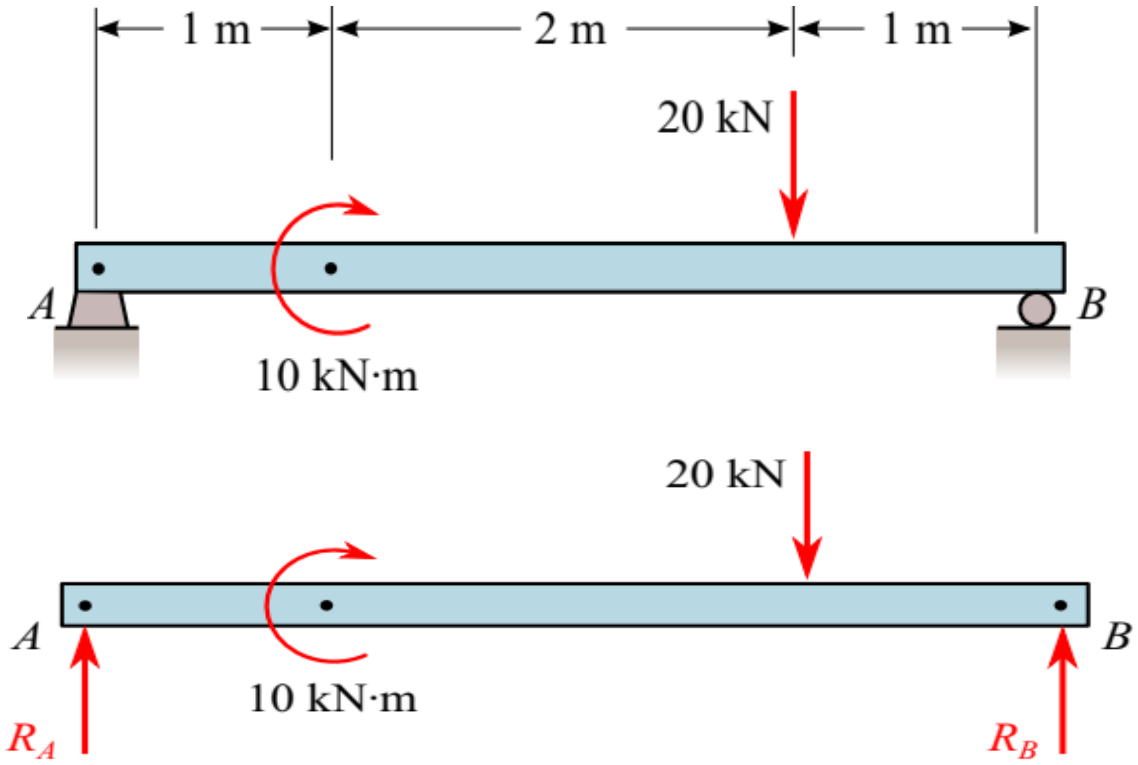


$$C_L = C_K = C = \frac{110}{2} = 55mm$$

$$\sigma_K = \sigma_{\min} = -\frac{Mc}{I} = -\frac{106{,}25.10^3.55}{3617500} = -1.625MPa$$

$$\sigma_L = \sigma_{\max} = \frac{Mc}{I} = \frac{106{,}25.10^3.55}{3617500} = 1.625MPa$$

Şekildeki gibi yüklenmiş ve mesnetlenmiş olan kirişin A mesnedinin 2 m sağındaki kesidinde ortaya çıkan kesme kuvveti V yi ve eğilme momenti M yi bulunuz.



$$\uparrow^{+} \sum F_y = R_A - 20 + R_B = 0$$

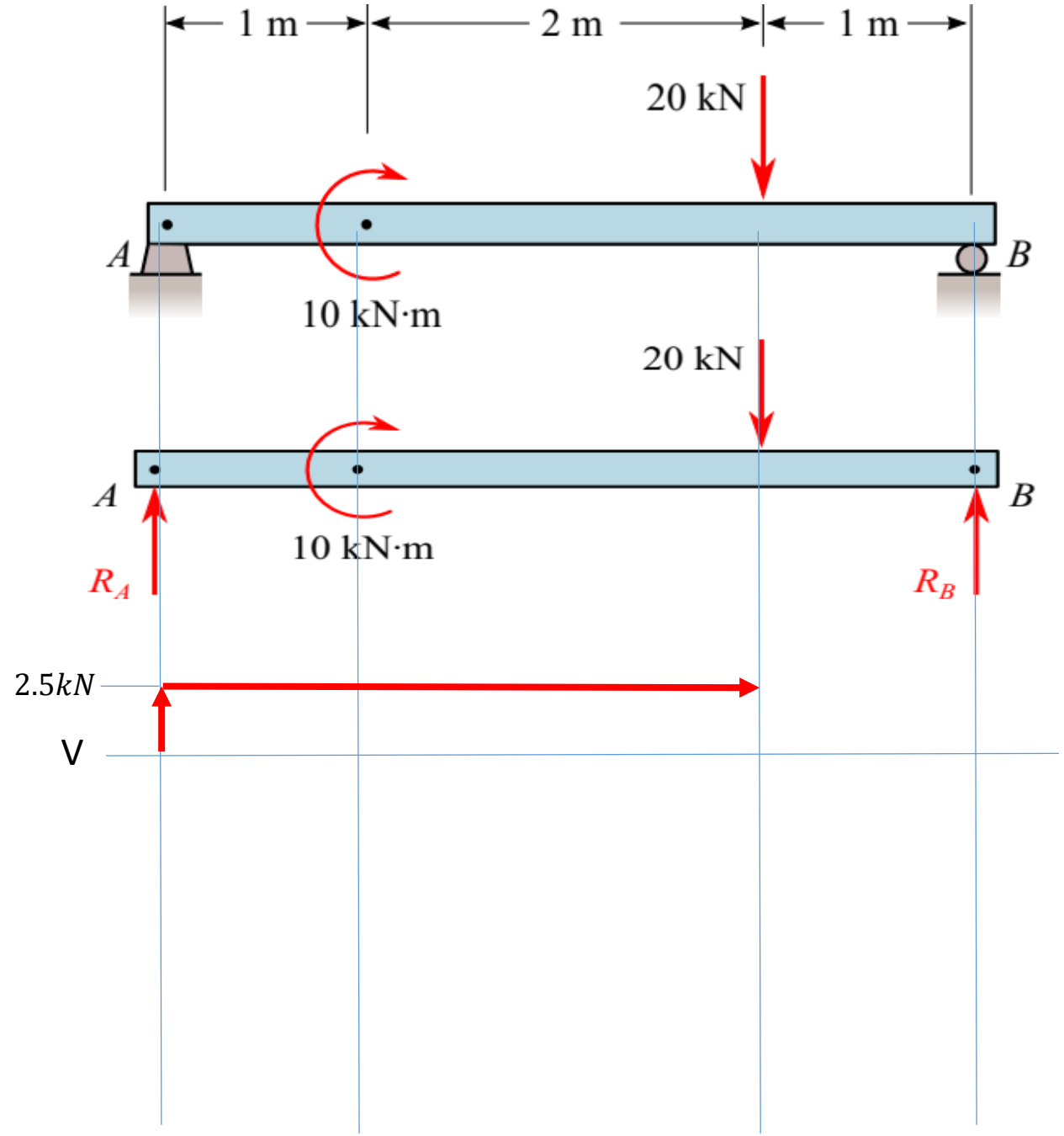
$$R_A + R_B = 20$$

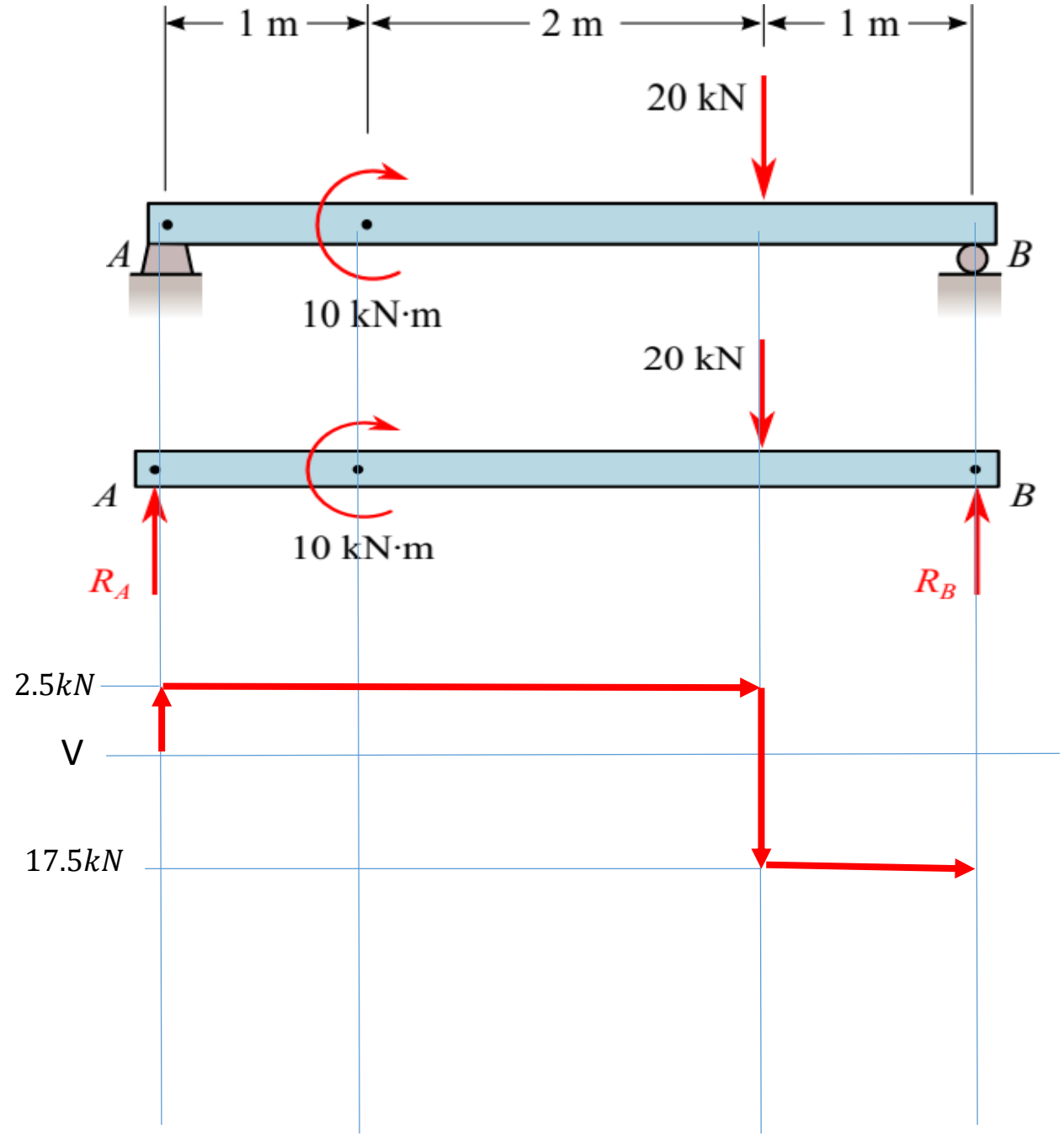
$$\sum M_A = -10 - 20.3 + R_B.4 = 0$$

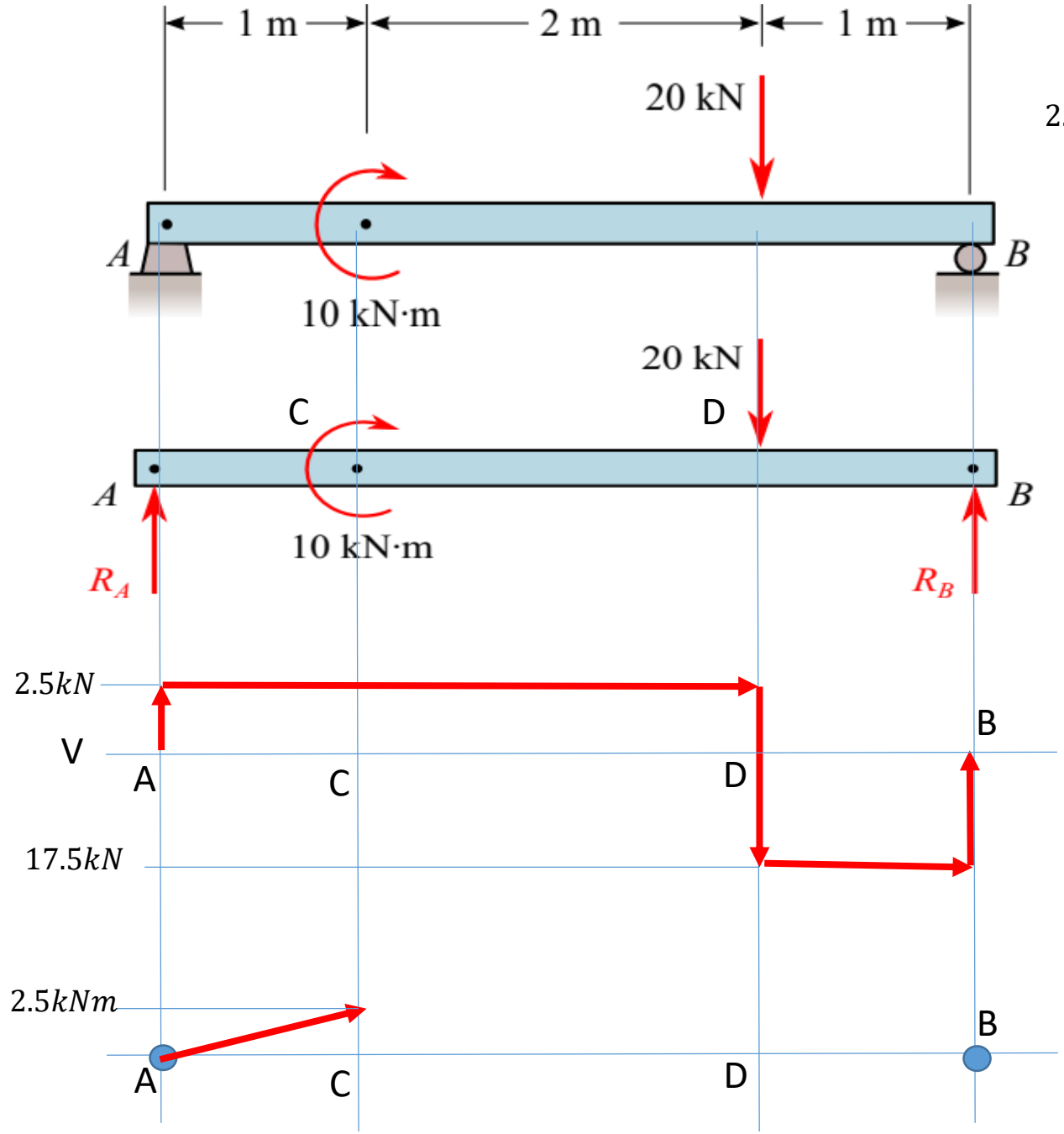
$$R_B = 17.5kN$$

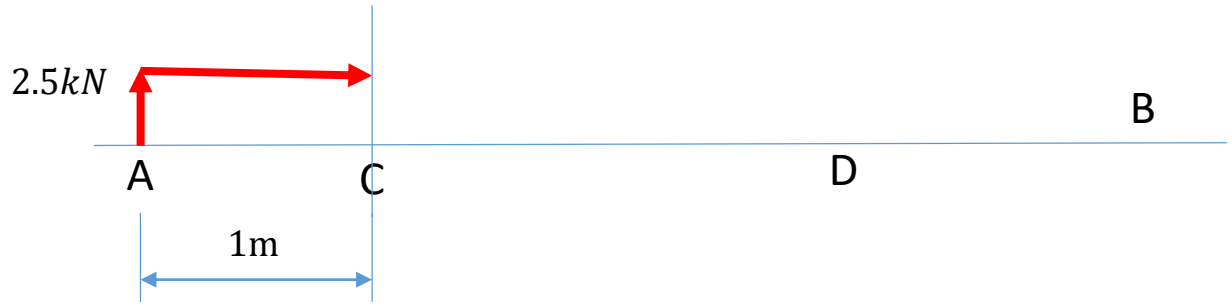
$$R_A + R_B = 20$$

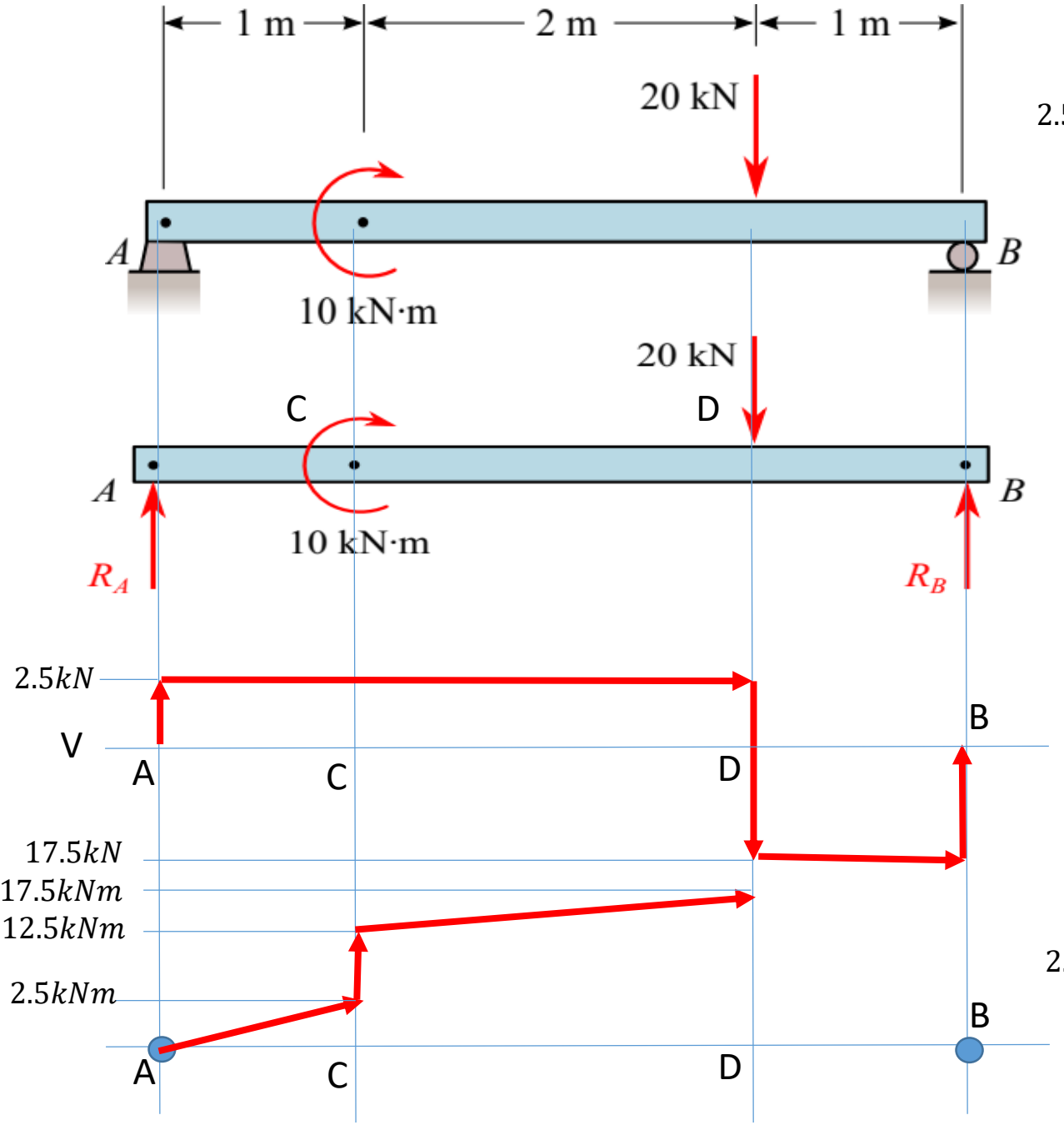
$$R_A = 2.5kN$$

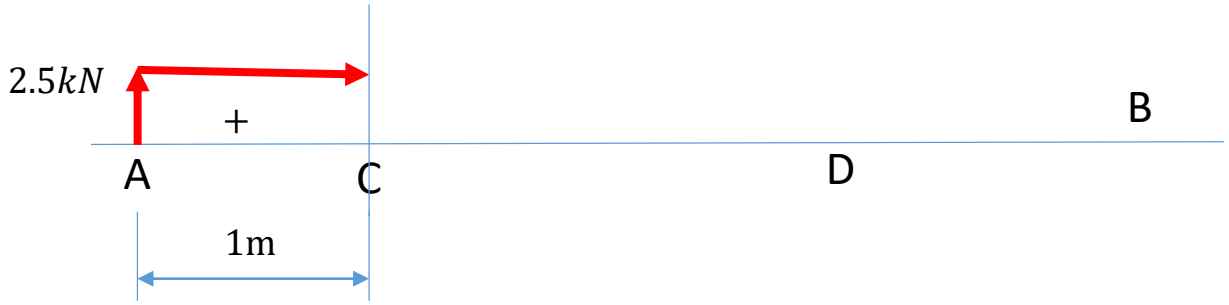

1 m
2 m
1 m
20 kN
A
B
10 kN·m
20 kN
A
B
10 kN·m
R_A
R_B
$2.5kN$
V

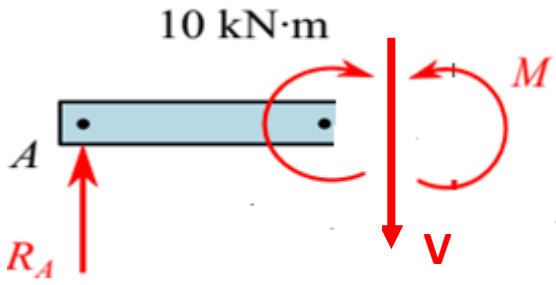

1 m
2 m
1 m
20 kN
A
B
10 kN·m
20 kN
A
B
10 kN·m
R_A
R_B
$2.5kN$
V
$17.5kN$


1 m
2 m
1 m
20 kN
A
B
10 kN·m
20 kN
C
D
A
B
10 kN·m
R_A
R_B
2.5kN
V
A
C
D
B
17.5kN
2.5kNm
A
C
D
B


2.5kN
A
C
D
B
1m



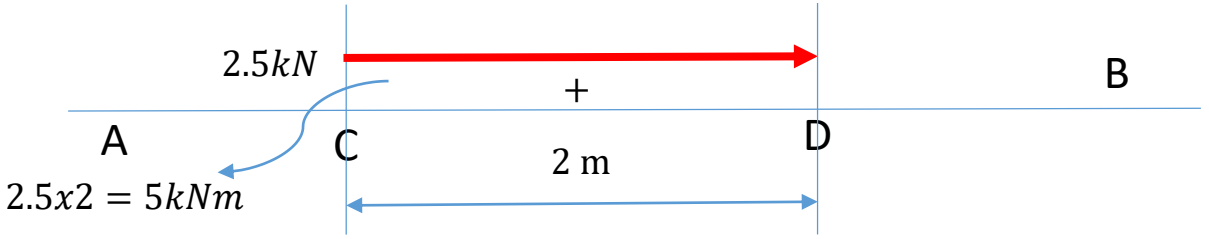




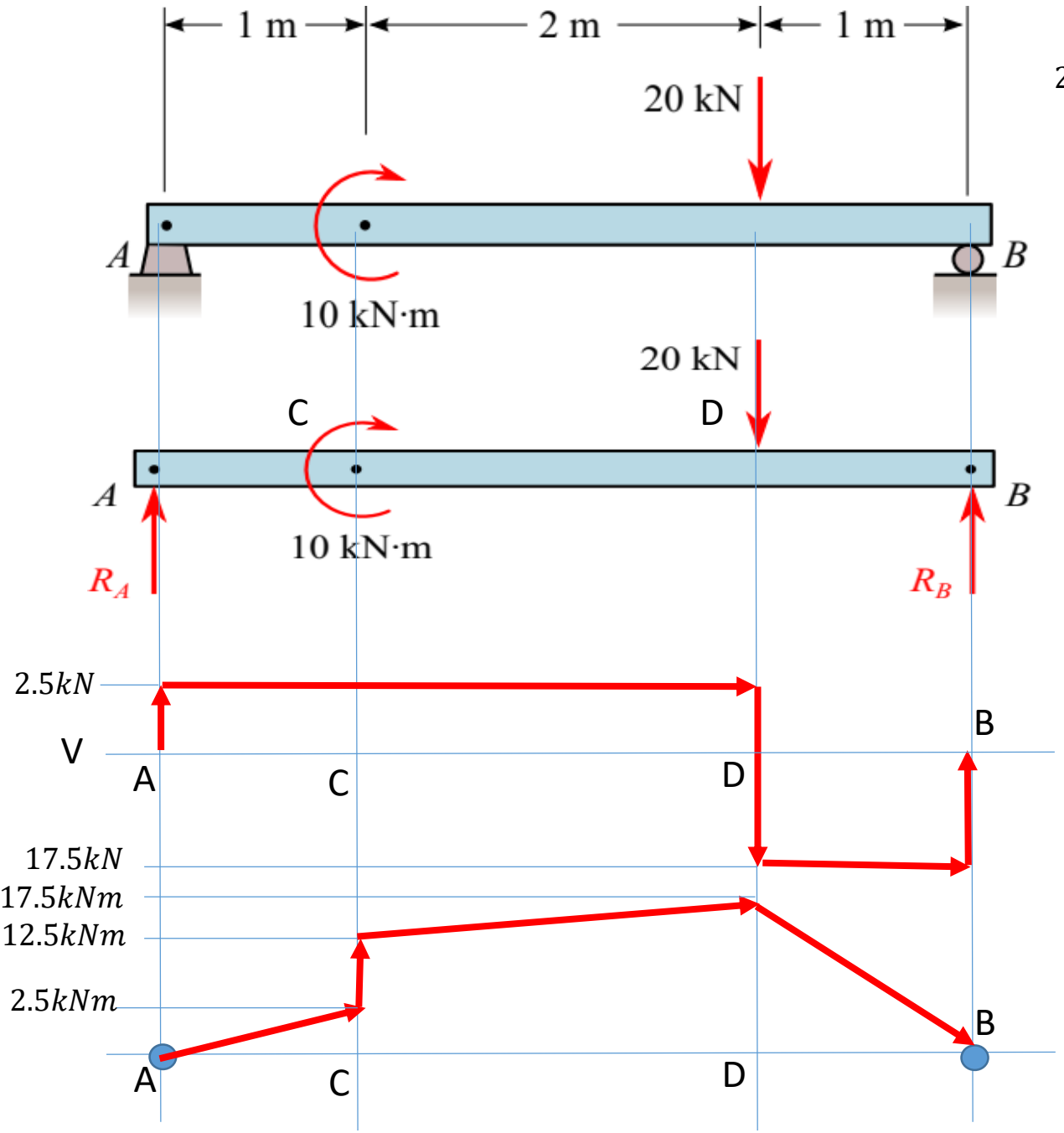
$$\sum M = 0$$

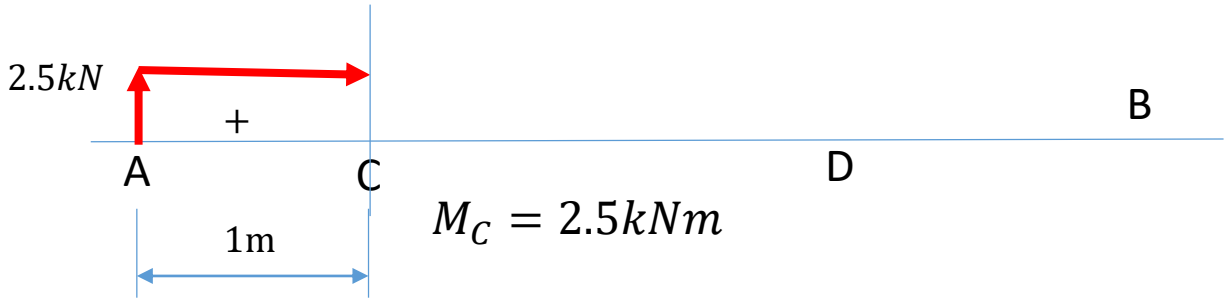
$$\sum M = M - 10 - 2.5x1 = 0$$

$$M = 12.5kNm$$

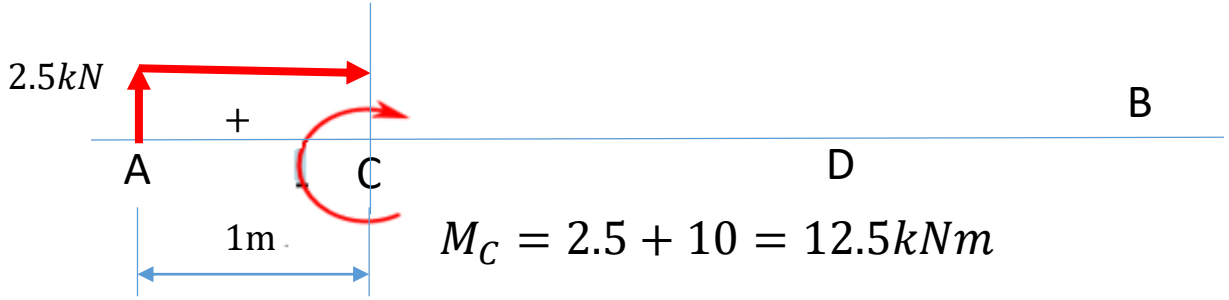


$$M_D = 12.5 + 5 = 17.5kNm$$

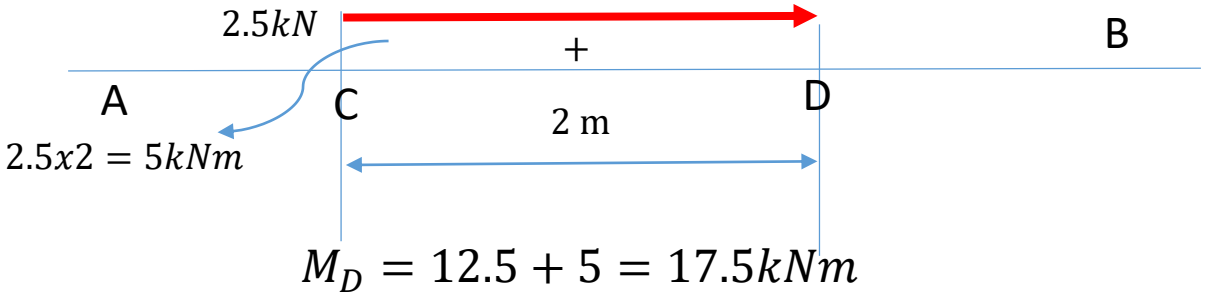




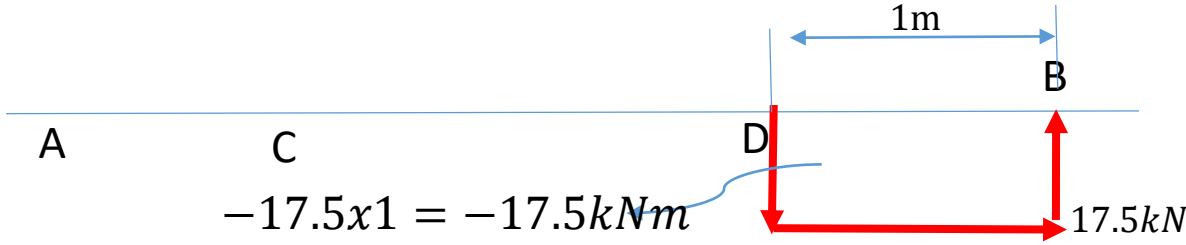
$M_C = 2.5kNm$



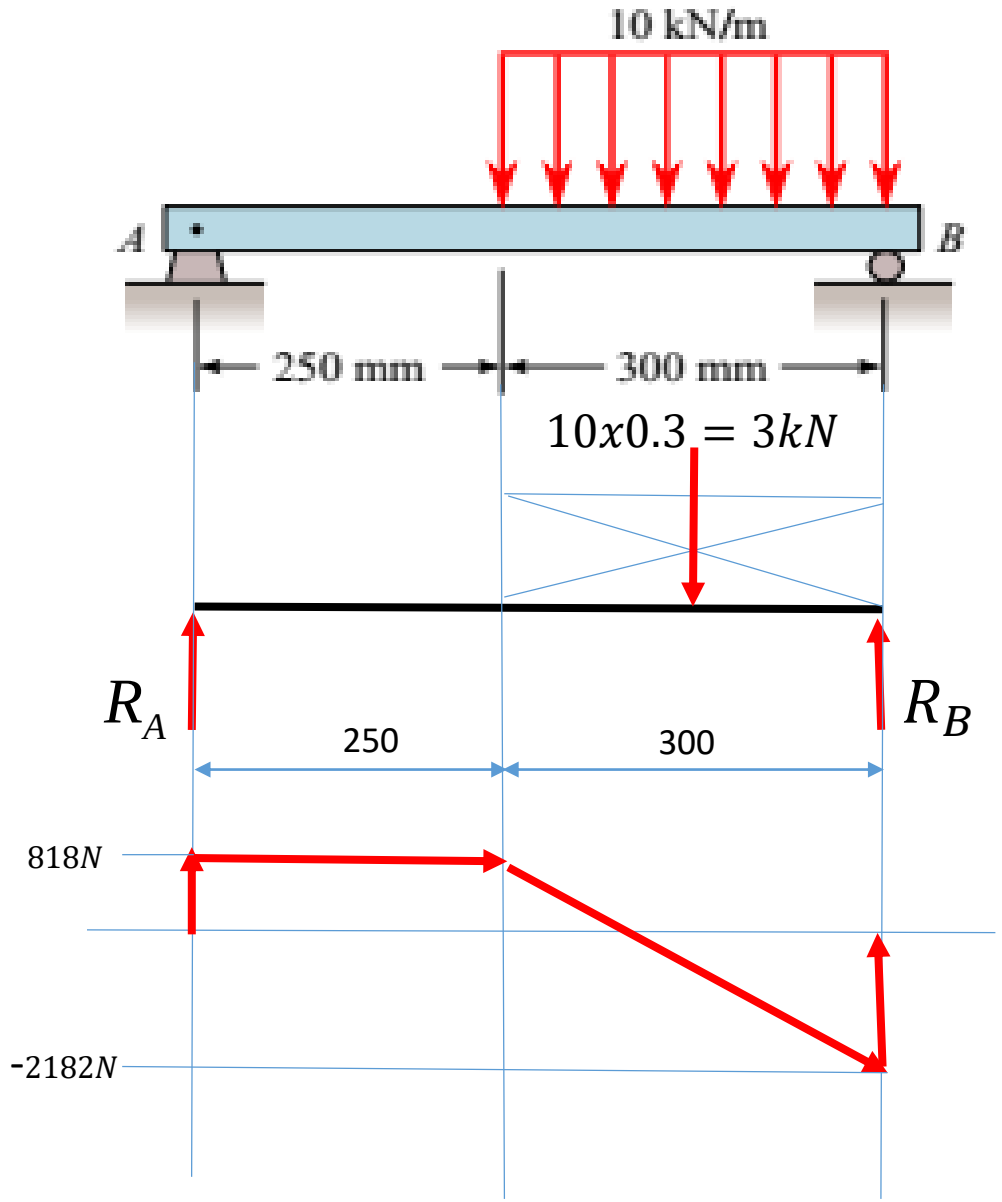
$M_C = 2.5 + 10 = 12.5kNm$



$M_D = 12.5 + 5 = 17.5kNm$

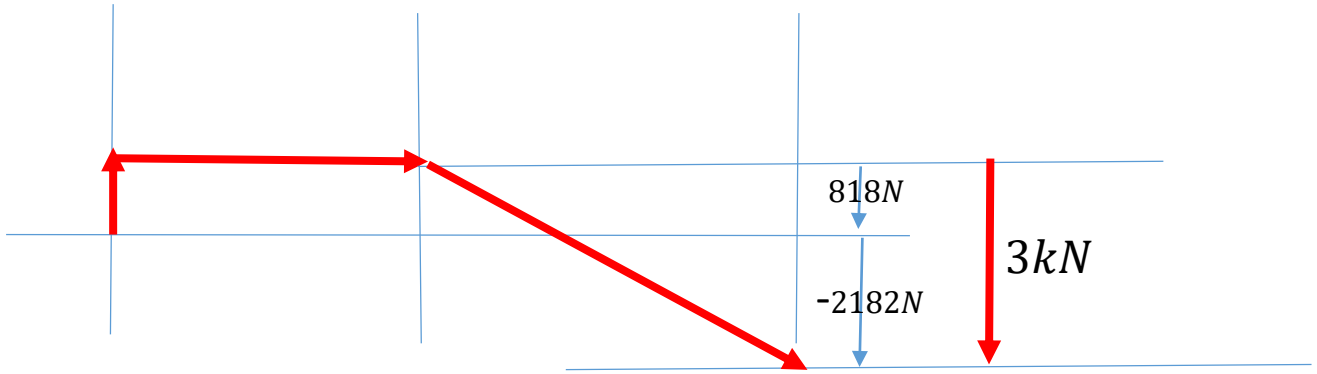


$M_B = -17.5 + 17.5 = 0$

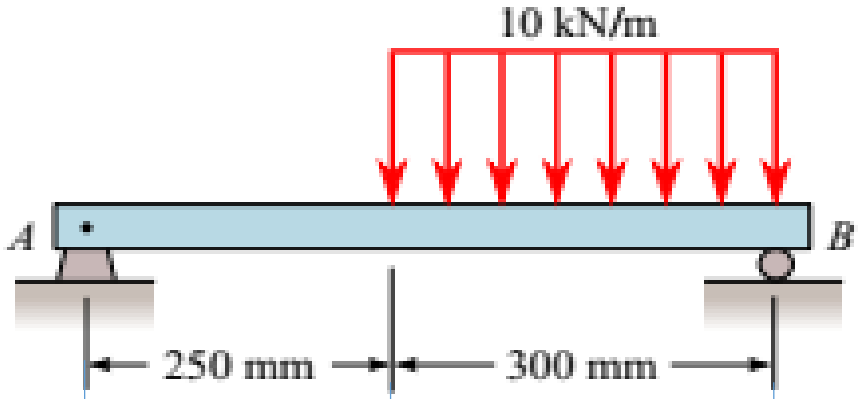


$$\uparrow \sum F_y = R_A - 3 + R_B = 0 \qquad R_A + R_B = 3$$

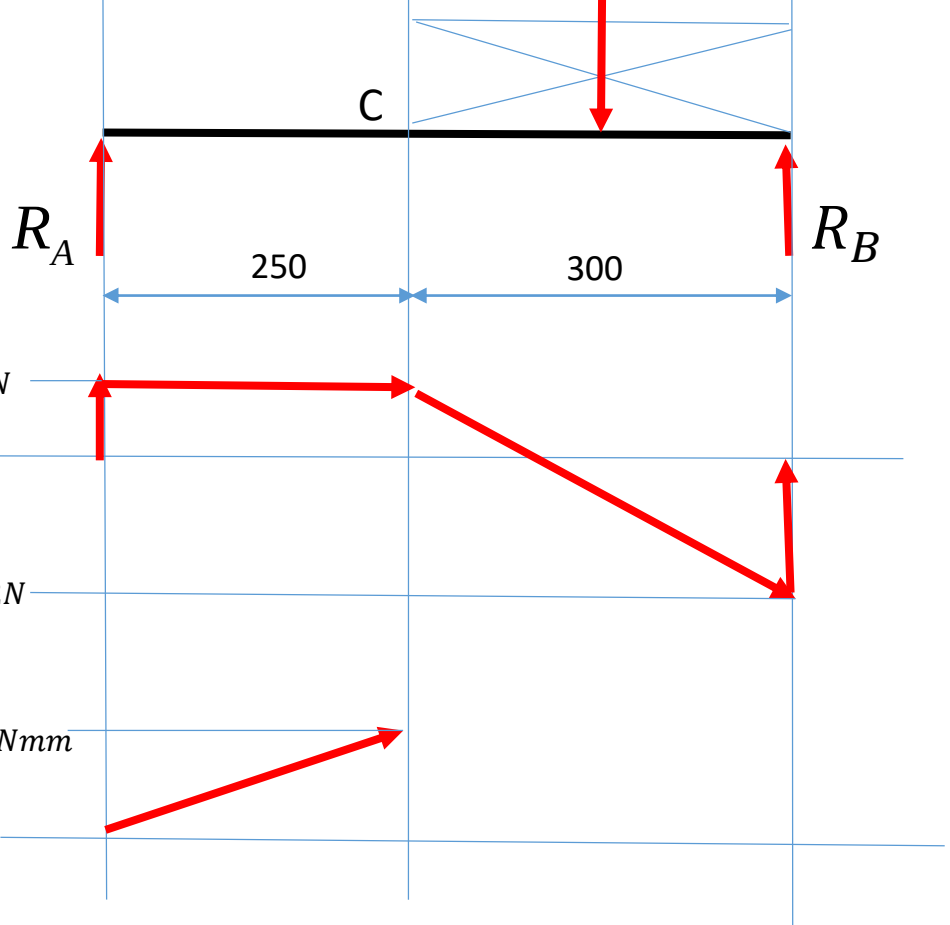
$$\sum M = -3x400 + R_B x550 = 0 \qquad R_B = 2182N \quad R_A = 818N$$

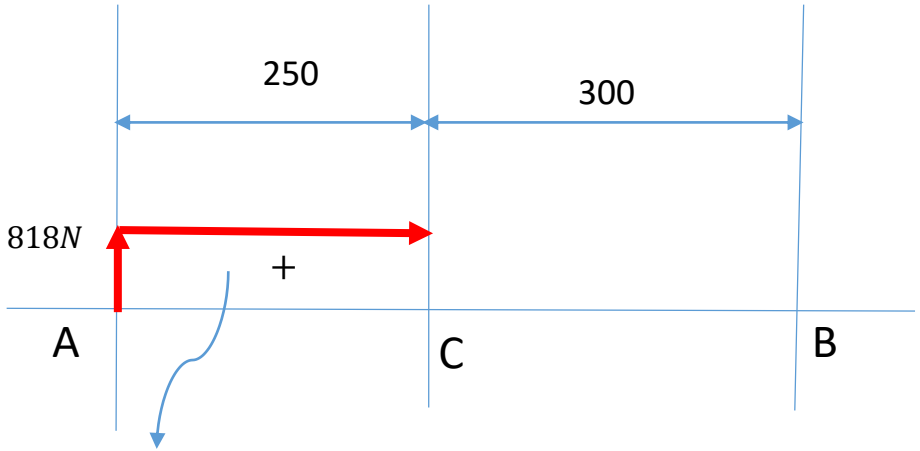


Üniform yayılı yükün bulunduğu kesit başlangıcı ve bitişi eğik çizgi İle birleşir. Yayılı yük aşağıya doğru ise aşağıya doğru, yukarıya doğru ise yukarıya doğru başlangıç ve bitişi arasındaki yükseklik yayılı kuvvetin bileşkesine eşit olacak şekilde çizilir.

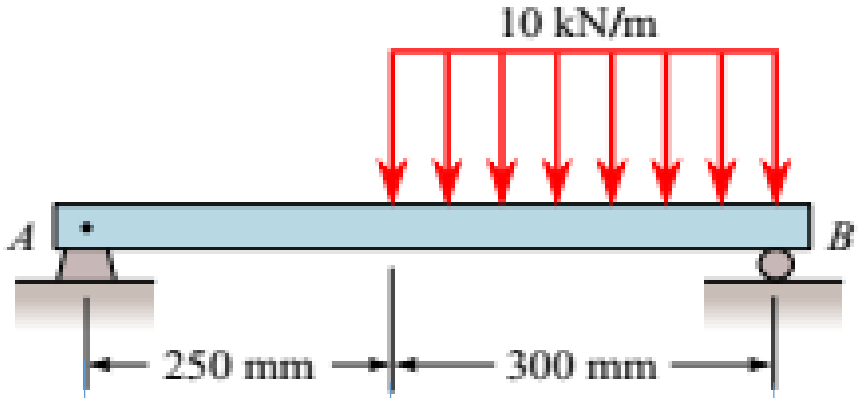


$10x0.3 = 3kN$

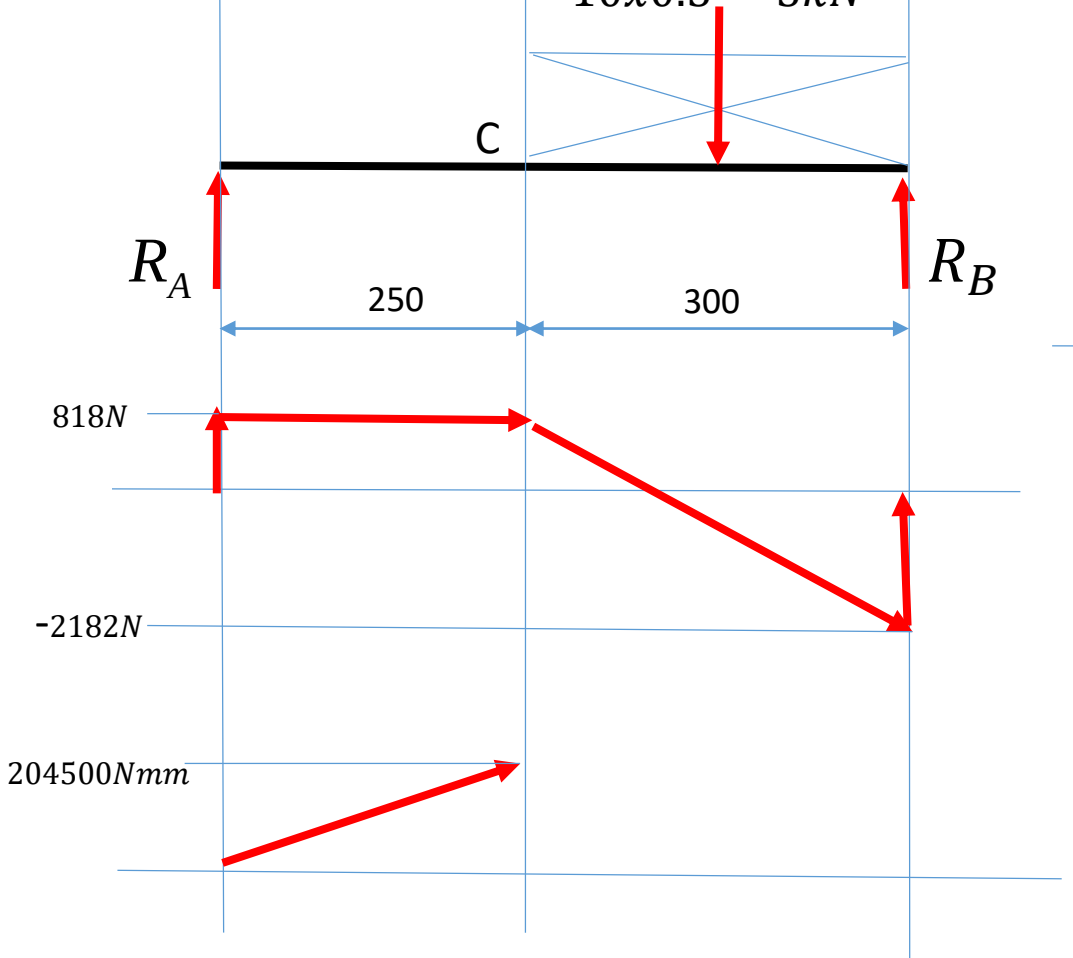


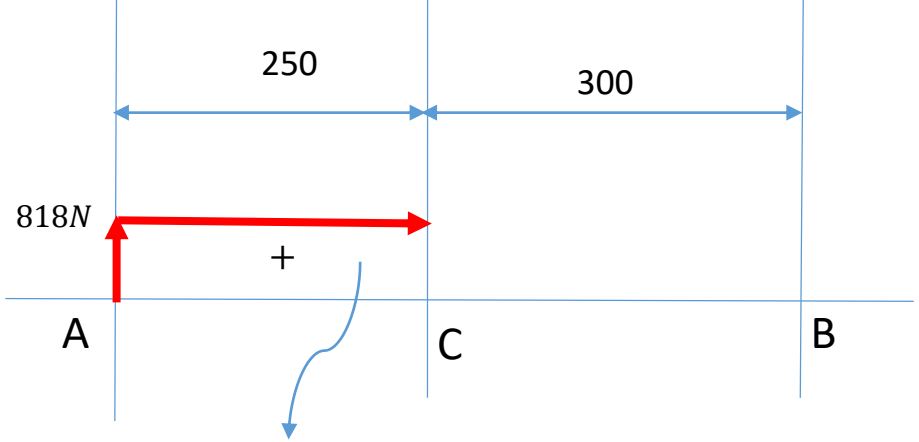


$818x250 = 204500Nmm$ $M_C = 204500Nmm$



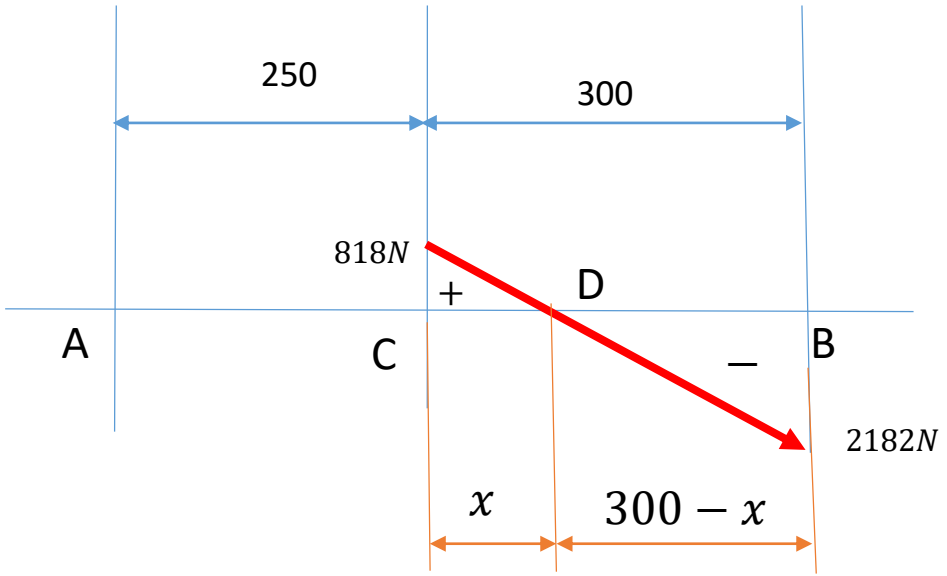
$10x0.3 = 3kN$





$818x250 = 204500Nmm$

$M_C = 204500Nmm$

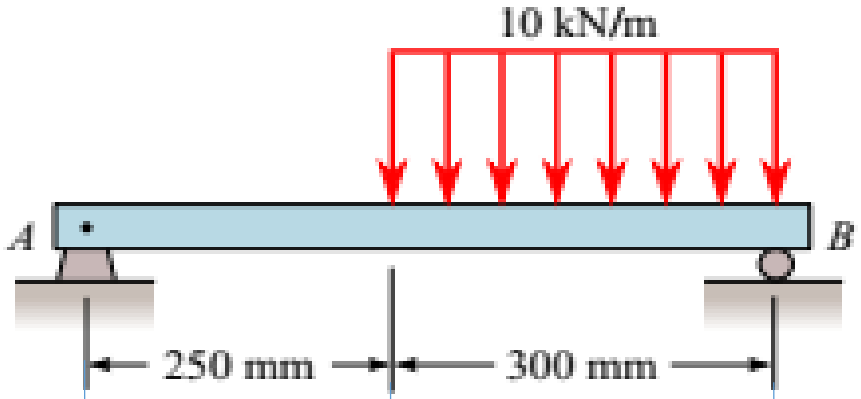


$$\frac{818}{2182} = \frac{x}{300 - x}$$

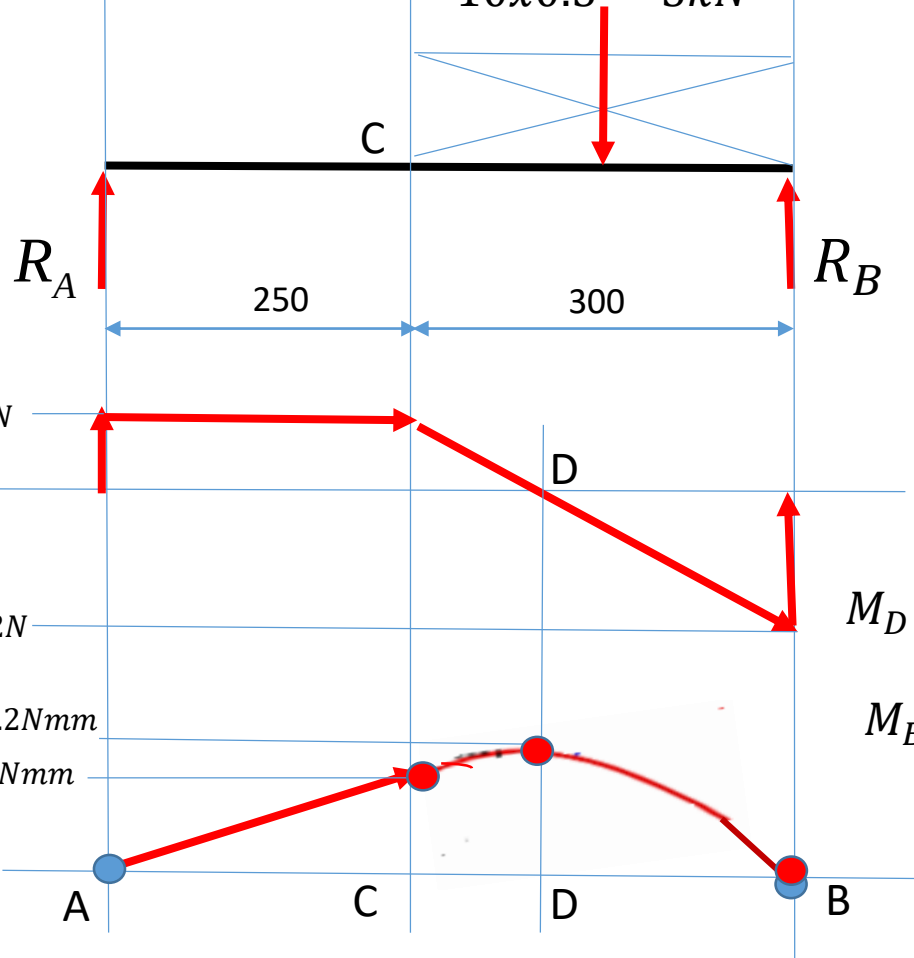
$$x = 81.8mm$$

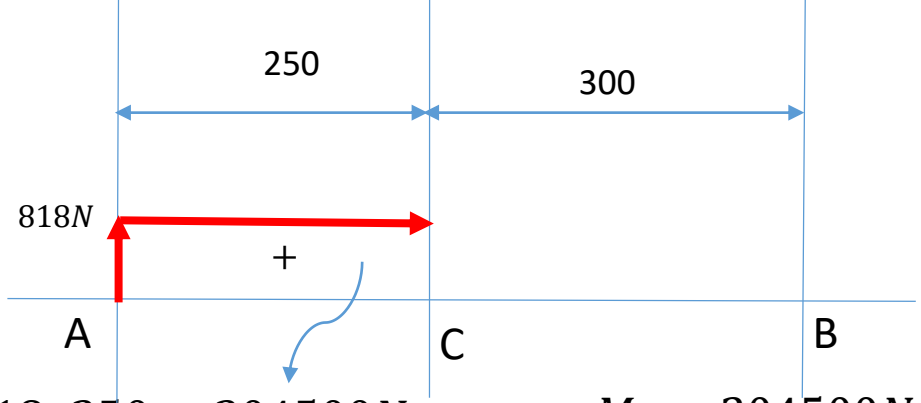
$$M_D = 204500 + \frac{81.8x818}{2} = 237956.2Nmm$$

$$M_B = 237956.2 - \frac{218.2x2182}{2} = 0$$

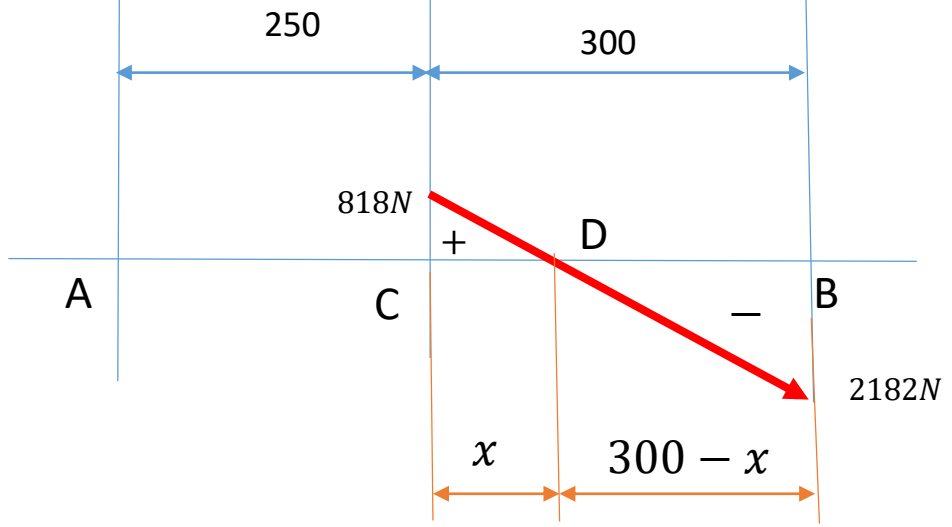


$10x0.3 = 3kN$





$818x250 = 204500Nmm$ $M_C = 204500Nmm$



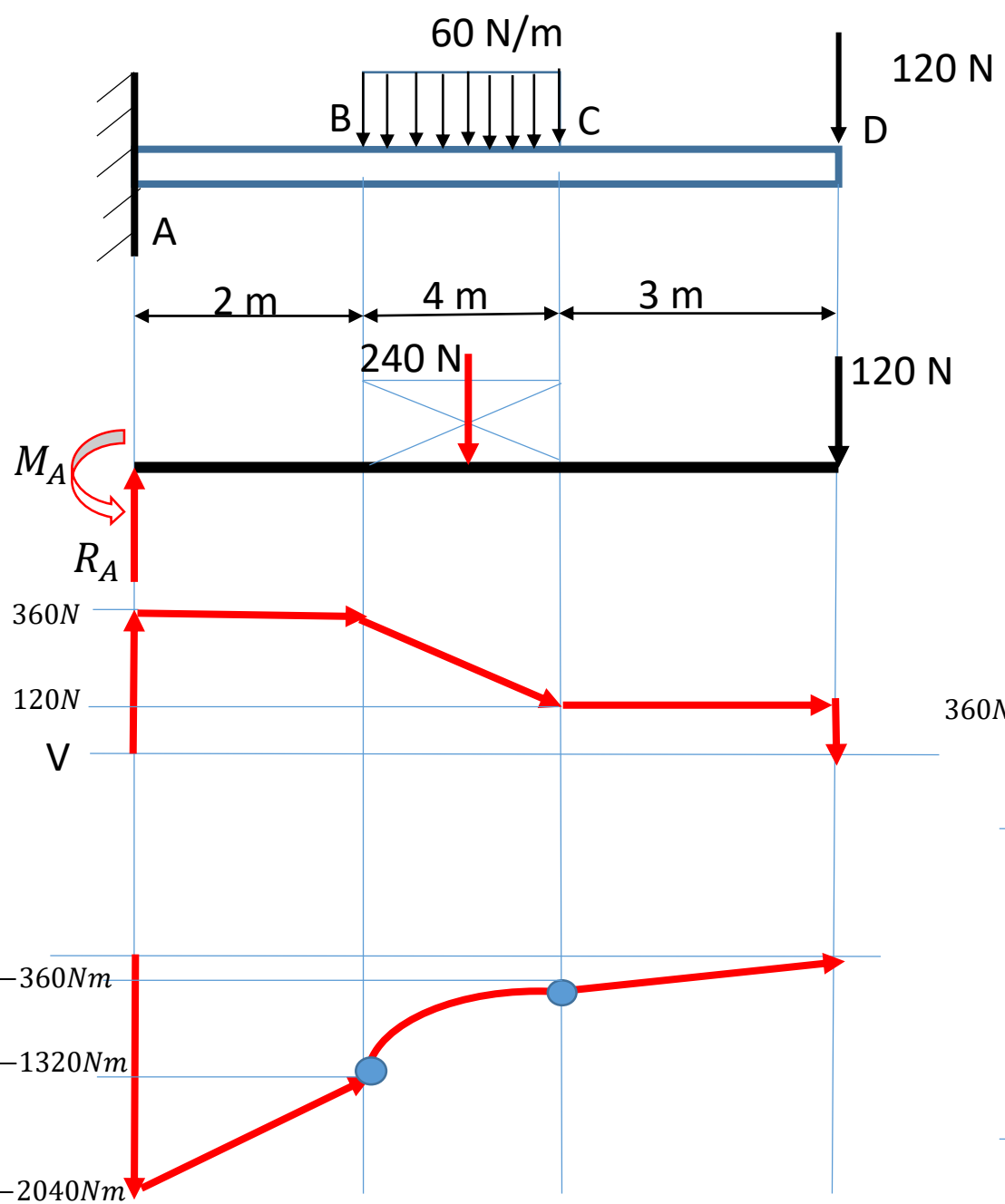
$$\frac{818}{2182} = \frac{x}{300 - x}$$

$$x = 81.8mm$$

$$M_D = 204500 + \frac{81.8x818}{2} = 237956.2Nmm$$

$$M_B = 237956.2 - \frac{218.2x2182}{2} = 0$$

Üniform yayılı yükün bulunduğu bölgede eğilme momenti diyagramı parabol çıkar. Hızlı ve kabaca çizim için eğri için 3 noktanın koordinatlarının belirlenmesi çizimi için yeterlidir.

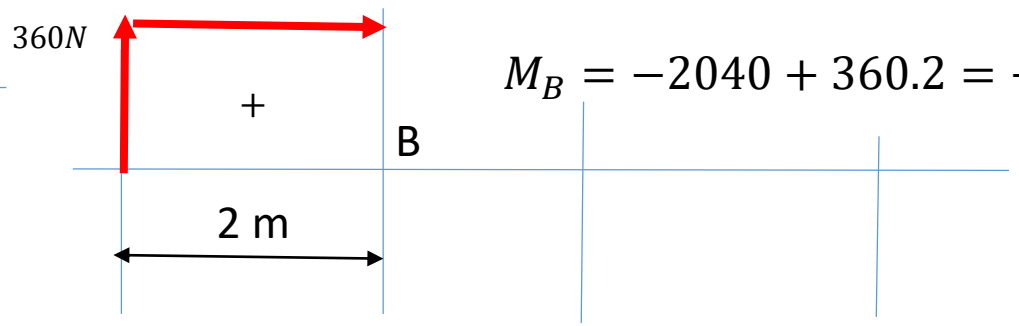


$$+\uparrow \sum F_y = R_A - 240 - 120 = 0 \qquad R_A = 360N$$

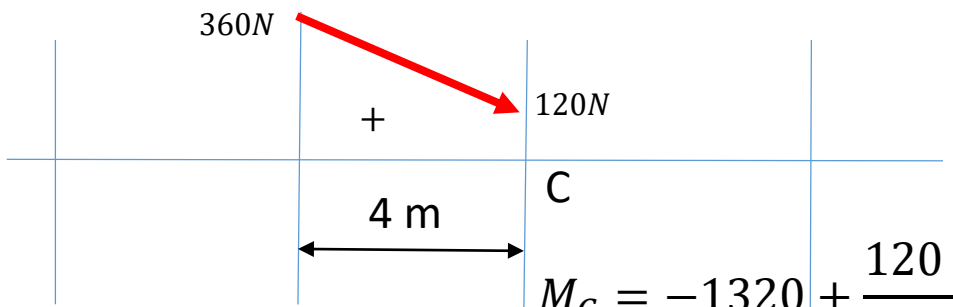
$$+\circlearrowleft \sum M_A = M_A - 240.4 - 120.9 = 0 \qquad M_A = 2040Nm$$



$$\sum M = M_A + M = 0 \qquad M = -M_A = -2040Nm$$

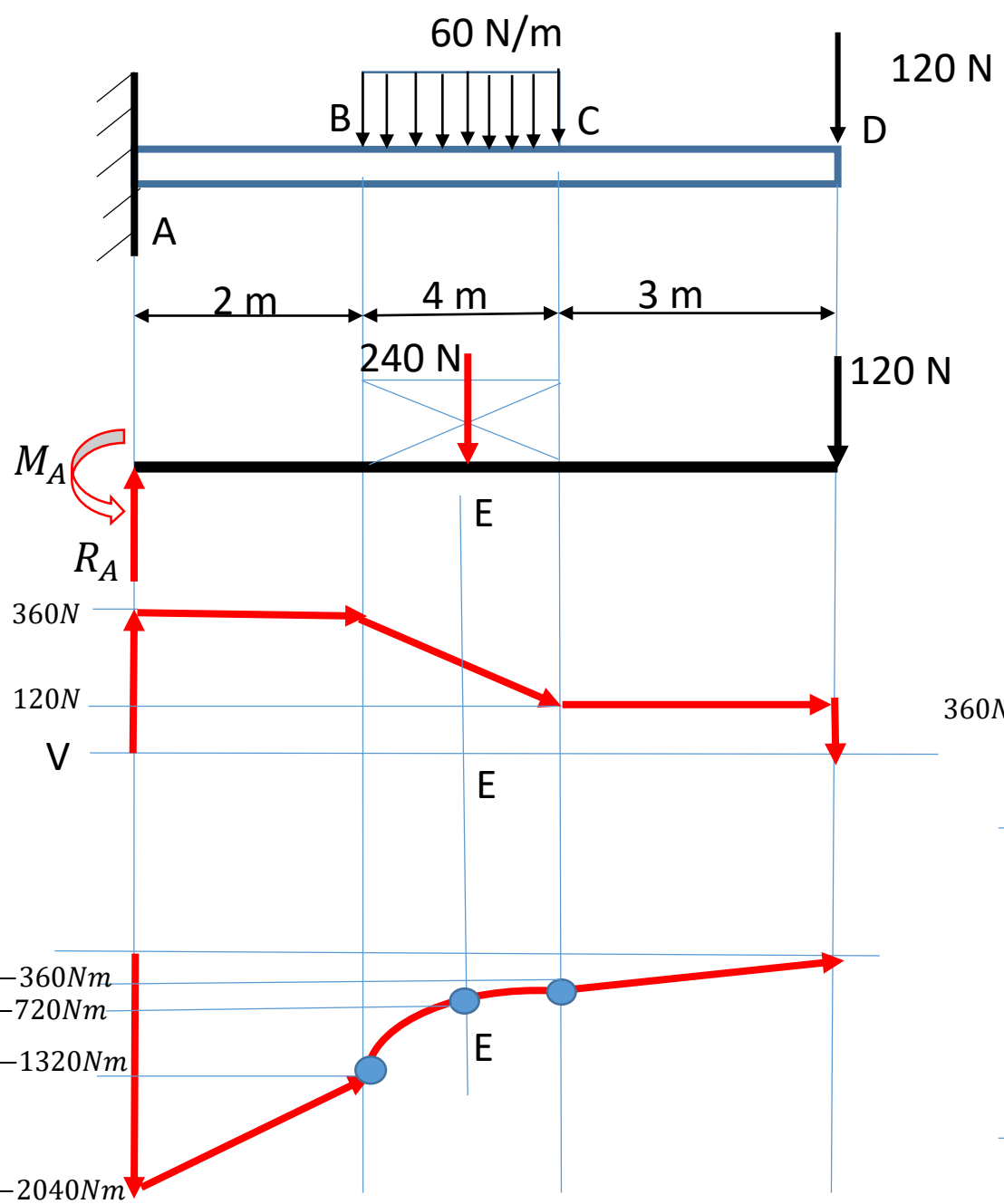


$$M_B = -2040 + 360.2 = -1320Nm$$



$$M_C = -1320 + \frac{120 + 360}{2} x4 = -360Nm$$

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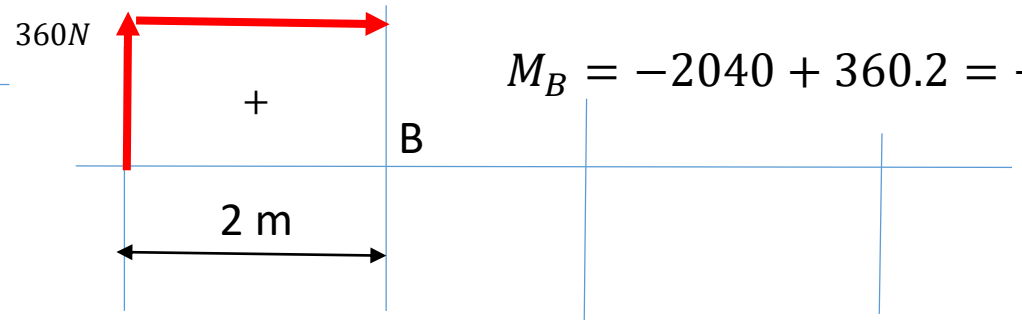


$$+\uparrow \sum F_y = R_A - 240 - 120 = 0 \qquad R_A = 360N$$

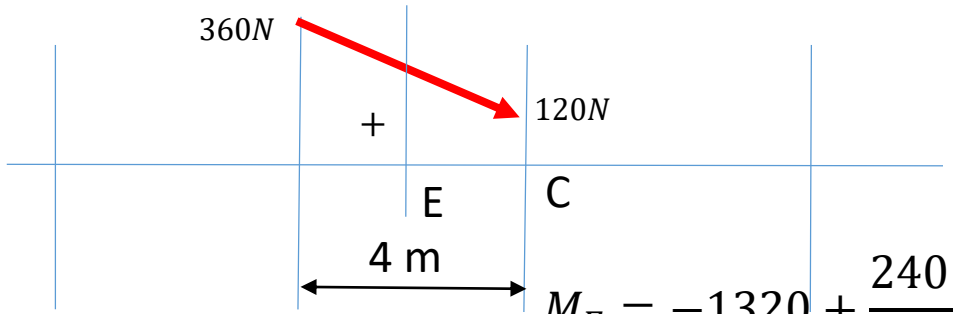
$$+\circlearrowleft \sum M_A = M_A - 240.4 - 120.9 = 0 \qquad M_A = 2040Nm$$



$$\sum M = M_A + M = 0 \qquad M = -M_A = -2040Nm$$



$$M_B = -2040 + 360.2 = -1320Nm$$



$$M_E = -1320 + \frac{240 + 360}{2} x2 = -720Nm$$



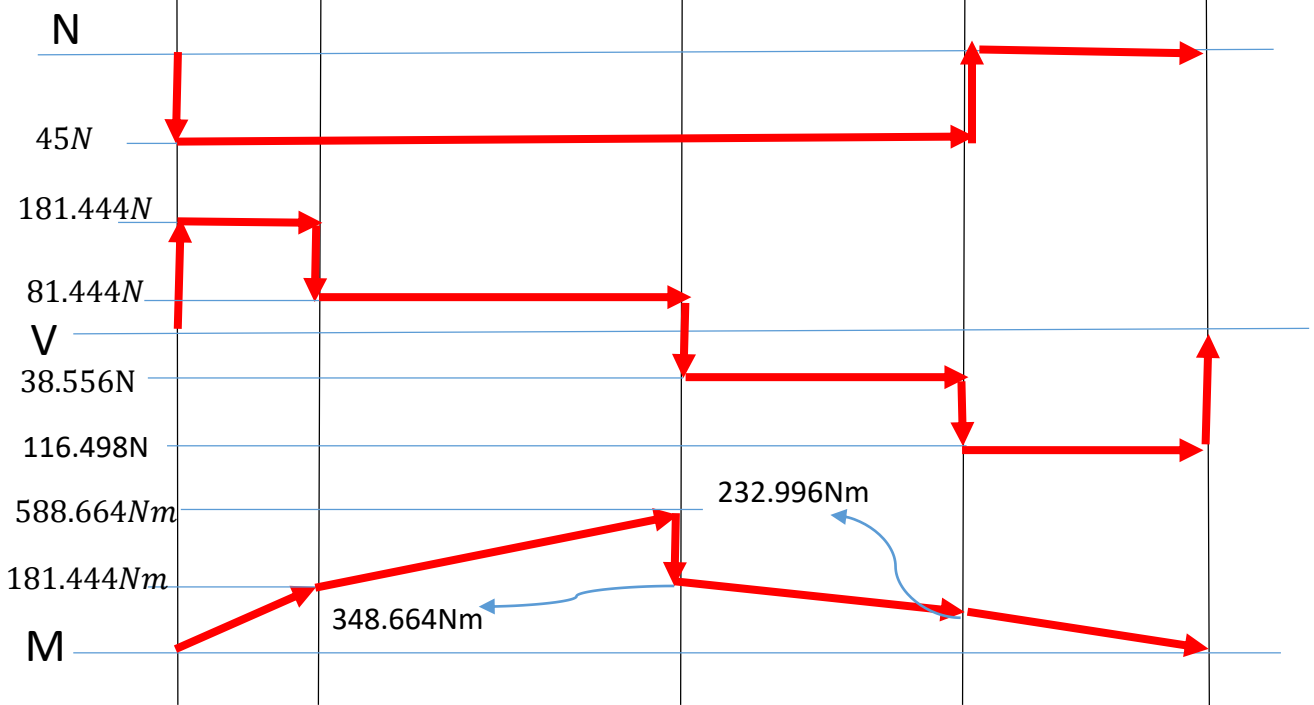
$$\sum F_x = R_{A_x} - \cos 60 x 90 = 0 \qquad R_{A_x} = 45N$$

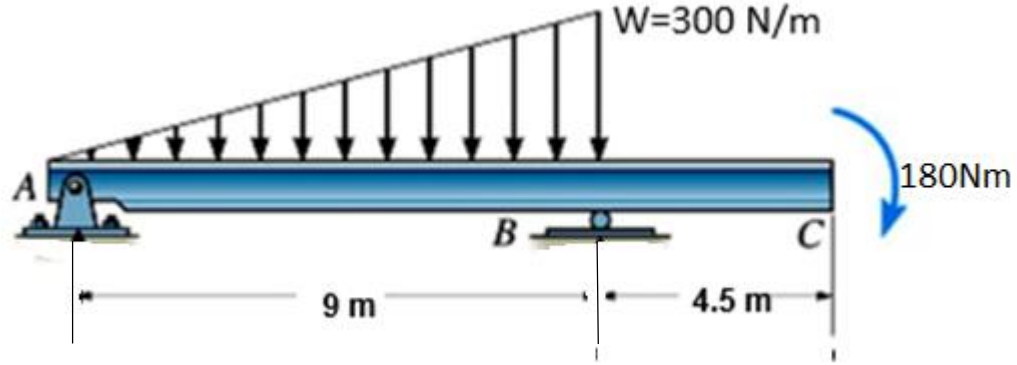
$$+\uparrow \sum F_y = R_{A_y} - 100 - 120 - \sin 60 x 90 + R_E = 0$$

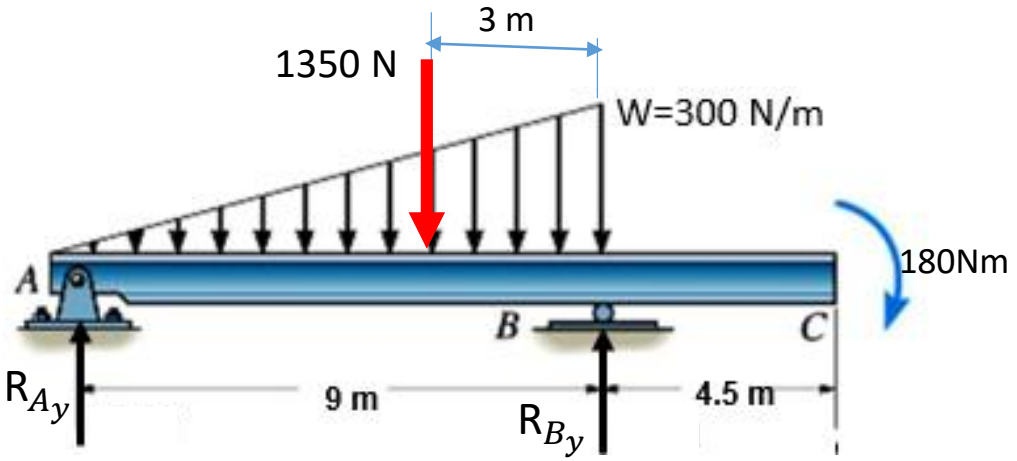
$$R_{A_y} + R_E = 297.942$$

$$\sum M_A = -100x1 - 120x6 + 240 - 90x\sin 60x9 + R_E x11 = 0$$

$$R_E = 116.498N \qquad R_{A_y} = 181.444N$$







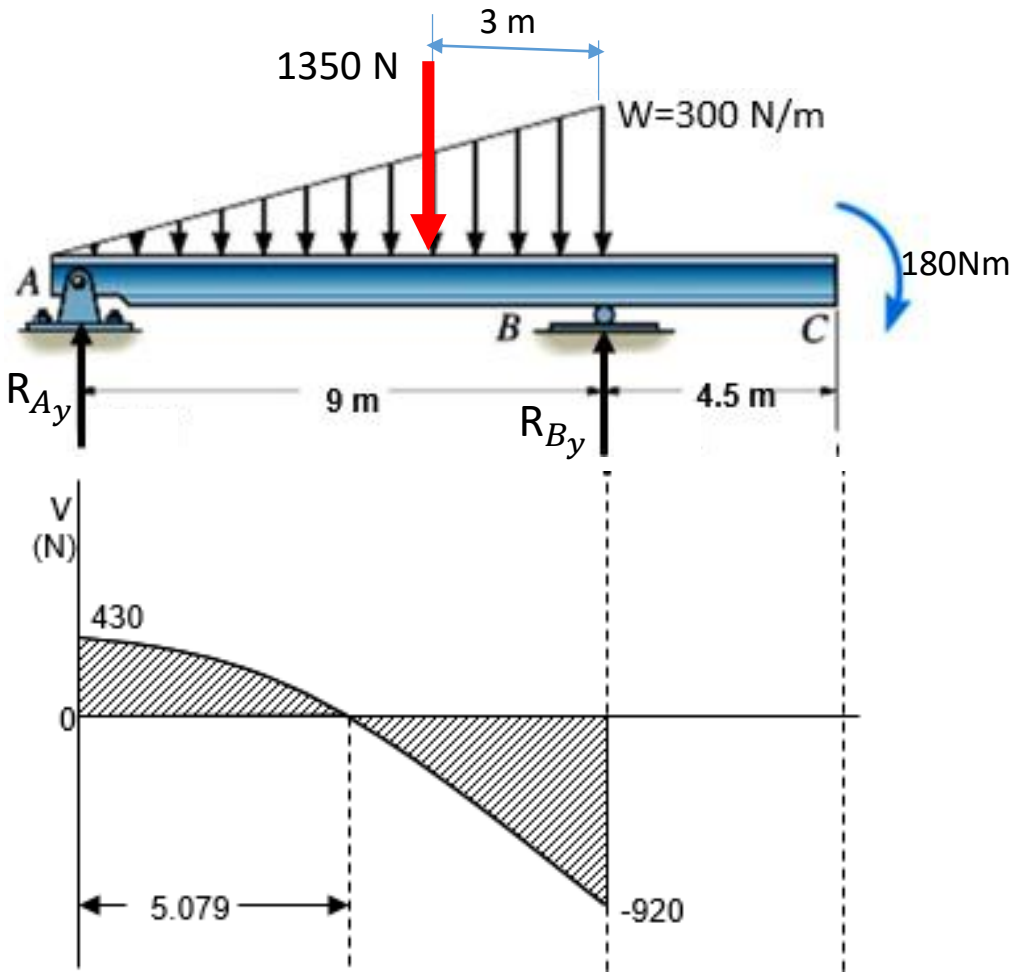
$$\Sigma M_A = 0 = R_{B_y}(9) - 180 - 0.5(9)(300)(6)$$

$$\Sigma M_B = 0 = -R_{A_y}(9) - 180 + 0.5(9)(300)(3)$$

$$R_{B_y} = 920\ N$$

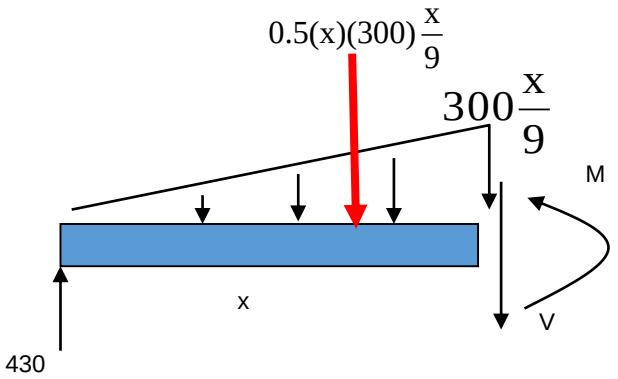
$$R_{A_y} = 430\ N$$

$$\Sigma Fy = 0 = R_{A_y} + R_{B_y} - 0.5(9)(300)$$



$$R_{B_y} = 920\ N$$

$$R_{A_y} = 430\ N$$

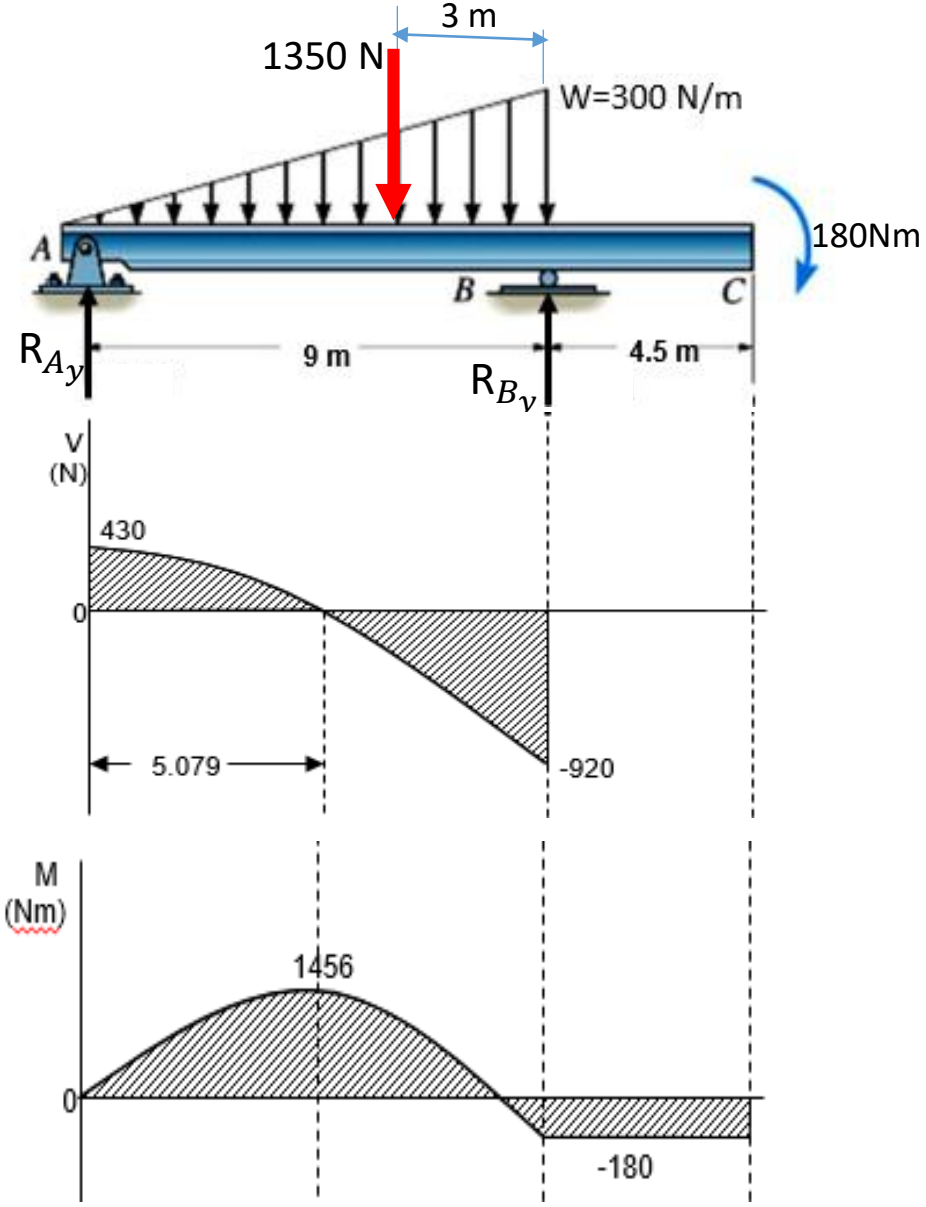


$$\sum F_y = -430 + 0.5(x)(300)\frac{x}{9} + V = 0$$

$$V = 430 - \frac{300\,x^2}{18} \quad V = 0 \quad x = 5.079\ \text{m}$$

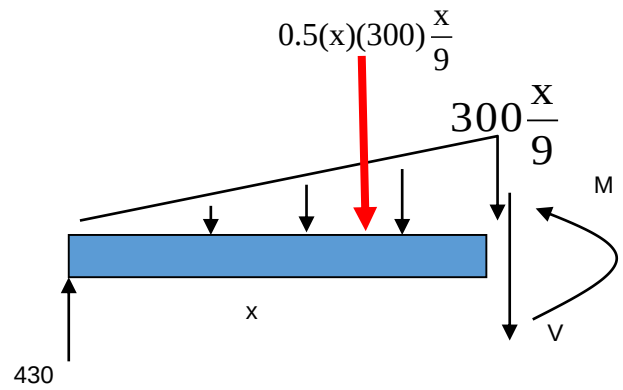
$$\Sigma M_{cut} = 0 = -430\,(x) + M + \left[0.5(x)(300)\frac{x}{9}\right]\left(\frac{x}{3}\right)$$

$$M = 430(x) - \frac{300\,x^3}{54}$$



$R_{B_y} = 920\ N$

$R_{A_y} = 430\ N$



$$\sum F_y = -430 + 0.5(x)(300)\frac{x}{9} + V = 0$$

$$V = 430 - \frac{300\,x^2}{18} \quad V = 0 \quad x = 5.079\ m$$

$$\Sigma M_{cut} = 0 = -430\,(x) + M + \left[0.5(x)(300)\frac{x}{9}\right]\left(\frac{x}{3}\right)$$

$$M = 430(x) - \frac{300\,x^3}{54}$$

$$M = 430.5{,}079 - \frac{300.5{,}079^3}{54} = 1456Nm$$

$$M = 430.9 - \frac{300.9^3}{54} = -180Nm$$